

Forcing a dense ideal on ω_1 over a model of determinacy

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Abstract

We force over a model of $\text{AD}_{\mathbb{R}} + (\Theta \text{ is regular})$ and arrive at a model in which $\text{ZFC} + \text{CH}$ holds and there exists a dense normal ideal on ω_1 ; in fact, this ideal will satisfy a little bit more. We show that in the same generic extension the non-stationary ideal on $[\mathbb{R}]^\omega$ is strong and pseudo-homogeneous. These notes are rewriting of [Sargsyan et al., 2021].

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[0.0.1] Properties of dense ideals.

[0.0.2] Prove: [3.2.4]

1 Prerequisites

1.1 Dependent Choice

[1.1.1] **Declaration.** We assume in this subsection that $V \models \text{ZF}$.

[1.1.2] **Definition.**

- a. Let λ be a limit ordinal. A λ -tree on X is a non-empty subset $T \subseteq X^{<\lambda}$ satisfying

$$(\forall s \in T)(\forall \alpha < \text{dom}(s))(s \restriction \alpha \in T).$$

- b. A tree is an ω -tree.

- c. $f : \gamma \rightarrow X$ is a *twig* of a λ -tree T on X if $\gamma < \lambda$ is a limit ordinal and

$$(\forall \alpha < \gamma)(f \restriction \alpha \in T).$$

- d. $f : \lambda \rightarrow X$ is a *branch* of a λ -tree T if

$$(\forall \alpha < \lambda)(f \restriction \alpha \in T).$$

- e. $s \in T$ is a *leaf* if it has no proper extension in T .

[1.1.3] **Definition.**

- a. $\kappa\text{-DC}_X$ is the following statement: every tree on X that has no leaves and contains all its twigs has a branch.

- b. $\kappa\text{-DC}$ is the statement

$$\forall X \kappa\text{-DC}_X.$$

- c. If $\kappa = \omega$, it is omitted from the notation.

[1.1.4] **Remark.** $\text{AC} \implies \forall \kappa \kappa\text{-DC}$.

[1.1.5] **Proposition.** If $\kappa < \lambda$, then

$$\lambda\text{-DC} \implies \kappa\text{-DC}.$$

Proof.

- (1) Let $T \subseteq X^{<\kappa}$ be such that
 - a. $(\forall s \in T)(\forall \alpha < \text{dom}(s))(s \restriction \alpha \in T)$
 - b. $(\forall s \in T)(\exists x \in X)(s \frown x \in T)$.
- (2) Let $T' \subseteq (X \cup \{*\})^{<\lambda}$ be such that $s \in T'$ if and only if either
 - $\text{dom}(s) < \kappa$ and $s \in T$,

or

 - $\text{dom}(s) \geq \kappa$, $s \restriction \alpha \in T$ for every $\alpha < \kappa$, and $s(\beta) = *$ for every $\beta \geq \kappa$.
- (3) T' satisfies conditions for $\lambda\text{-DC}$, so there is $f : \lambda \rightarrow X$ such that $f \restriction \alpha \in T'$ for every $\alpha < \lambda$.
- (4) $f \restriction \kappa$ then witnesses $\kappa\text{-DC}$ for T .

□

[1.1.6] **Proposition.** Assume that there is a surjection $F : X \times \text{Ord} \rightarrow V$ and suppose that $\lambda\text{-DC}_X$ holds. Then $\lambda\text{-DC}$ holds.

Proof.

- (1) Let T be a κ -tree with no leaves that contains all its twigs.
- (2) Let us define a λ -tree S on X by recursion on $\text{dom}(s)$ for $s \in S$, together with a family $(\alpha_s : s \in S)$ of ordinals.

- a. We assert $\emptyset \in S$.
 - b. Given $s \in S$, let $t_s := (F(s(i), \alpha_{s \upharpoonright i}) : i < \text{dom}(s))$. We inductively assume that $t_s \in T$ and that T/t_s has no leaves.
 - c. Let α_s be the minimal ordinal α such that there exists $x \in X$ with $t_s \frown F(x, \alpha) \in T$ and $T/(t_s \frown F(x, \alpha))$ has no leaves.
 - d. We assert that $s \frown x \in S$ if and only if $t_s \frown F(x, \alpha_s) \in T$. It is clear that the inductive assumptions remain satisfied.
 - e. At limit stage $\gamma < \lambda$, we set assert that $f : \gamma \rightarrow X$ belongs to S if and only if $f \upharpoonright \alpha \in S$ for every $\alpha < \gamma$.
- (3) λ -tree S on X has no leaves and it contains all its twigs. By DC_X , we conclude that it has a branch $f : \lambda \rightarrow X$.
- (4) It is now clear that

$$(F(f(\alpha), \alpha_{f \upharpoonright \alpha}) : \alpha < \lambda)$$

is a branch through T .

□

[1.1.7] **Lemma.** Assume $V \models \kappa\text{-DC}$. Let $\mathbb{P}$ be a κ -closed poset, let $g \rightsquigarrow V^{\mathbb{P}}$, and let $f \in (\kappa \rightarrow V)^{V[g]}$. Then $f \in V$.

Proof.

- (1) Suppose $p \Vdash \dot{f} : \kappa \rightarrow V$ and let us find $q \leq p$ and $g : \kappa \rightarrow V$ such that $q \Vdash \dot{f} = g$.
- (2) Let $T \subseteq \mathbb{P}^{<\kappa}$ consist of all $s \in \mathbb{P}^{<\kappa}$ such that
 - a. s is $\leq_{\mathbb{P}}$ -decreasing,
 - b. $s(0) = p$,
 - c. $s(\alpha + 1) \Vdash \dot{f}(\alpha)$ whenever $\alpha < \text{dom}(s)$.
- (3) By κ -closure, T satisfies conditions of $\kappa\text{-DC}$. Let $u : \kappa \rightarrow \mathbb{P}$ be such that $u \upharpoonright \alpha \in T$ for every $\alpha < \kappa$.
- (4) Let q be such that $q \leq u(\alpha)$ for every $\alpha < \kappa$.
- (5) Let $f : \kappa \rightarrow V$ be defined by

$$g(\alpha) = x \iff q \Vdash \dot{f}(\alpha) = x.$$

- (6) Then q, g are as required.

□

[1.1.8] **Proposition.** Let M be an inner model for $\text{ZF} + \text{DC}_X$ and let $T \in M$ be a tree on X . If T has branch in V , then it has a branch in M .

1.2 Forcing and Topology

[1.2.1] **Definition.** Let $\mathbb{P}$ be a poset. Subset $g \subseteq \mathbb{P}$ is a *filter* if

- a. $1_{\mathbb{P}} \in g$
- b. $p \in g, p \leq q \implies q \in g$
- c. $p, q \in g \implies (\exists r \in g)(r \leq p, q)$.

A filter is *maximal* if it is maximal w.r.t. $\subseteq$.

[1.2.2] **Definition.** We define $\mathcal{F}_{\mathbb{P}} := \{g \subseteq \mathbb{P} : g \text{ is a maximal filter}\}$. The default topology on this set is the one whose base is given by

$$\mathcal{N}_p := \{g \in \mathcal{F}_{\mathbb{P}} : p \in g\} \quad (p \in \mathbb{P}).$$

[1.2.3] **Remark.**

- a. It is easily verified that this is indeed a base for a topology.
- b. If $\mathbb{P} = \text{Col}(\omega, X)$, then $\mathcal{F}_{\mathbb{P}} \approx X^\omega$.
- c. If $\mathbb{P} = \mathbb{B} - \{0_{\mathbb{B}}\}$, where $\mathbb{B}$ is a boolean algebra, then $\mathcal{F}_{\mathbb{P}}$ is the Stone space of $\mathbb{B}$.

[1.2.4] **Definition.** Poset $\mathbb{P}$ is *nice* if it has the following property: if $p, q \in \mathbb{P}$ have a lower bound, then they have the¹ infimum (denoted $p \cdot q$).

[1.2.5] **Remark.** In the case of a nice poset, property **c** of [1.2.1] is equivalent to saying

$$(\forall p, q \in g)(p \parallel q \wedge p \cdot q \in g).$$

[1.2.6] **Lemma.** Suppose that $\mathbb{P}$ is a nice poset. Then for every $C \subseteq \mathbb{P}$ satisfying

$$(\forall p, q \in C)(p \parallel q),$$

there exists $g \in \mathcal{F}_{\mathbb{P}}$ satisfying $C \subseteq g$.

Proof.

- (1) Let $F := \{q : n < \omega, p_0, \dots, p_{n-1} \in C, q \geq \prod_{i < n} p_i\}$. Set F is a filter in $\mathbb{P}$.
- (2) By Zorn's lemma, there exists a maximal filter $g \supseteq F \supseteq C$.

□

[1.2.7] **Proposition.** Let $\mathbb{P}$ be a nice poset. Then:

- a. $\mathcal{N}_p$ is clopen for every $p \in \mathbb{P}$;
- b. $\mathcal{F}_{\mathbb{P}}$ is Hausdorff;
- c. $\mathcal{F}_{\mathbb{P}}$ is compact.

Proof.

- (1) For **a**, note that if $g \notin \mathcal{N}_p$, then by [1.2.6] there exists $q \in g$ with $q \perp p$. This means $g \in \mathcal{N}_q$ and $\mathcal{N}_q \cap \mathcal{N}_p = \emptyset$.
- (2) For **b**, note that if $g \neq h$, then by maximality there are $p \in g - h$ and $q \in h - g$.
- (3) By [1.2.6], we have $p^* \in h$ with $p^* \leq q$ and $p \perp p^*$ and $q^* \in g$ with $q^* \leq p$ and $q \perp q^*$.
- (4) This means:
 - a. $\mathcal{N}_{p^*} \ni h$ and $\mathcal{N}_{q^*} \ni g$;
 - b. $(\forall k \in \mathcal{N}_{p^*})(k \not\geq p)$
 - c. $(\forall k \in \mathcal{N}_{p^*})(k \not\geq q^*)$
 - d. $(\forall k \in \mathcal{N}_{q^*})(k \not\geq q)$
 - e. $(\forall k \in \mathcal{N}_{q^*})(k \not\geq p^*)$
 - f. $\mathcal{N}_{p^*} \cap \mathcal{N}_{q^*} = \emptyset$.
- (5) For **c**, given a family $(\mathcal{N}_{p_i} : i \in I)$ of basic clopen sets with finite intersection property, use [1.2.6] to get $g \in \mathcal{F}_{\mathbb{P}}$ with $g \supseteq \{p_i : i \in I\}$.
- (6) We have $g \in \bigcap_{i \in I} \mathcal{N}_{p_i}$.

□

[1.2.8] **Remark.** In particular, $\mathcal{F}_{\mathbb{P}}$ satisfies Baire's Category Theorem.

¹We implicitly assume that posets are orders (and not just quasi-orders), so an infimum is indeed unique.

[1.2.9] **Lemma.** Suppose that $\mathbb{P}$ is a nice poset. Then for every pre-dense set $D \subseteq \mathbb{P}$, set

$$U_D := \bigcup_{p \in D} \mathcal{N}_p = \{q \in \mathcal{F}_{\mathbb{P}} : (\exists p \in D)(p \in g)\}$$

is open and dense.

Proof.

- (1) Openness is obvious. Let us verify density.
- (2) Let $q \in \mathbb{P}$ be arbitrary and let us show $\mathcal{N}_q \cap U_D \neq \emptyset$.
- (3) Since D is pre-dense, there is $p \in D$ such that $p \parallel q$.
- (4) By [1.2.6], there is $g \in \mathcal{F}_{\mathbb{P}}$ with $g \ni p, q$.
- (5) Then $g \in \mathcal{N}_p \subseteq U_D$ as well as $g \in \mathcal{N}_q$.

□

[1.2.10] **Proposition.** If M is a countable transitive model of set theory and $\mathbb{P} \in M$ is a nice poset, then the set of all $g \rightsquigarrow M^{\mathbb{P}}$ is co-meager in $\mathcal{F}_{\mathbb{P}}$.

Proof.

- (1) For $A \subseteq \mathbb{P}$ a maximal anti-chain, U_A is open and dense.
- (2) For $A \subseteq \mathbb{P}$ a maximal anti-chain and $g \in \mathcal{F}_{\mathbb{P}}$, we have $g \cap A \neq \emptyset$ if and only if $g \in U_A$.
- (3) Thus, $g \rightsquigarrow M^{\mathbb{P}}$ if and only if $g \cap A \neq \emptyset$ for every maximal anti-chain $A \in M$ in $\mathbb{P}$, if and only if

$$g \in \bigcap \{U_A : A \in M \text{ a maximal anti-chain in } \mathbb{P}\}.$$

- (4) This last set is a countable intersection of dense open sets, i.e. co-meager.

□

1.3 Properties of Ideals

[1.3.1] **Definition.** Let F be a filter on $[X]^{\omega}$.

- a. F is *fine* if

$$(\forall x \in X)(\{a \in [X]^{\omega} : x \in a\} \in F).$$

- b. F is *normal* if it is fine and for every $S \in F^+$ and every

$$f : S \rightarrow [X]^{\omega}$$

satisfying

$$(\forall a \in S)(\emptyset \subset f(a) \subseteq a),$$

there exist $x \in X$ and $S' \in F^+$ with $S' \subseteq S$ such that

$$(\forall a \in S')(x \in f(a)).$$

[1.3.2] **Definition.** For $F : X \rightarrow \mathcal{P}([X]^{\omega})$, we define

$$\Delta_{x \in X} F(x) := \{a \in [X]^{\omega} : a \in \bigcap_{x \in a} F(x)\}.$$

[1.3.3] **Proposition.** Fine filter D on $[X]^{\omega}$ is normal if and only if for every $F : X \rightarrow D$, we have $\Delta F \in D$.

Proof.

Implication ($\Rightarrow$)

- (1) Let $F : X \rightarrow D$ and assume towards contradiction that $\nabla F \notin D$.
(2) We have $S := (\nabla F)^{\mathbb{C}} \in D^+$ and we can define

$$f : S \rightarrow [X]^\omega : a \mapsto \{x \in a : a \notin F(x)\}.$$

- (3) We have $(\forall a \in S)(\emptyset \subset f(a) \subseteq a)$, so we conclude that there are $x \in X$ and $S' \in D^+$ with $S' \subseteq S$ and

$$(\forall a \in S')(x \in f(a)).$$

- (4) This means $S' \cap F(x) = \emptyset$, which is a contradiction.

Implication ($\Leftarrow$)

- (5) Let $S \in D^+$ and $f : S \rightarrow X$ with

$$(\forall a \in S)(\emptyset \subset f(a) \subseteq a)$$

be arbitrary.

- (6) We want to show

$$(\exists x \in X)(\{a \in S : x \in f(a)\} \in D^+).$$

Assume otherwise.

- (7) $(\forall x \in X)(\{a \in [X]^\omega : a \not\subseteq S \vee x \notin f(a)\} \in D)$
(8) Let $F : X \rightarrow D$ be defined by setting

$$F(x) := \{a \in [X]^\omega : a \not\subseteq S \vee x \notin f(a)\}.$$

We have $\nabla F \in D$.

- (9) In particular, there is $a \in \nabla F \cap S$.
(10) We have

$$a \in \bigcap_{x \in a} F(x).$$

Since $a \in S$, we have

$$(\forall x \in a)(x \notin f(a)).$$

- (11) This contradicts $\emptyset \subset f(a) \subseteq a$.

□

[1.3.4] **Exercise.** Assume that AC holds. Then normality of D is equivalent to its usual formulation: for every $S \in D^+$ and every

$$f : S \rightarrow X$$

satisfying

$$(\forall a \in S)(f(a) \in a),$$

there exist $x \in X$ and $S' \in D^+$ with $S' \subseteq S$ such that

$$(\forall a \in S')(f(a) = x).$$

[1.3.5] **Definition.** Set $A \subseteq [X]^\omega$ is *club* if there exists $F : X^{<\omega} \rightarrow X$ such that

$$\{M \in [X]^\omega : F[M^{<\omega}] \subseteq M\} \subseteq A.$$

[1.3.6] **Proposition.** Club sets form a normal filter. □

[1.3.7] **Definition.** Let M be an inner model for ZF such that $M^\omega \subseteq M$ and let I be an ideal on $[\mathbb{R}]^\omega$. We say that ideal I has *the ordinal covering property w.r.t. M* if

$$(\forall f : [\mathbb{R}]^\omega \rightarrow \text{Ord})(\forall S \in I^+)(\exists f' \in ([\mathbb{R}]^\omega \rightarrow \text{Ord}) \cap M)(\exists S' \in I^+ : S' \subseteq S)(f \restriction S' = f' \restriction S').$$

[1.3.8] **Remark.** The above notion is an auxiliary one which will play a role in our main arguments.

[1.3.9] **Declaration.** For the rest of this section, we work under the assumption that $V \models \text{ZFC}$.

[1.3.10] **Notation.** For I an ideal on X , we denote by $\mathbb{P}_I$ the poset $(\mathcal{P}(X) - I, \subseteq)$. This poset is forcing equivalent to boolean algebra $\mathcal{P}(X)/I$.

[1.3.11] **Lemma.** Let I be a normal ideal on $[\mathbb{R}]^\omega$ and let $G \rightsquigarrow V^{\mathbb{P}_I}$. Then G is V -normal.

Proof. The usual argument. $\square$

[1.3.12] **Proposition.** Let I be a normal precipitous ideal on $[\mathbb{R}]^\omega$, let $G \rightsquigarrow V^{\mathbb{P}_I}$, and let $j := j_G : V \prec \text{Ult}(V, G) =: \mathfrak{M}$. Then $\text{crit}(j) = \omega_1^V$ and $j(\omega_1^V) \geq \mathfrak{c}^{+V}$.

Proof.

- (1) $\text{crit}(j) \geq \omega_1^V$ is obvious.
- (2) For a function $f \in V$ on $([\mathbb{R}]^\omega)^V$, we have:

$$[f] \in [\text{id}] \iff \{\sigma : f(\sigma) \in \sigma\} \in G \quad (1)$$

$$\iff (\exists x \in \mathbb{R}^V)(\{\sigma : f(\sigma) = x\} \in G) \quad (2)$$

$$\iff (\exists x \in \mathbb{R}^V)([f] = j(x)) \quad (3)$$

$$\iff (\exists x \in \mathbb{R}^V)([f] = x). \quad (4)$$

(2) follows from [1.3.11].

- (3) Thus, $\mathbb{R}^V = j[\mathbb{R}^V] = [\text{id}] \in \mathfrak{M}$.
- (4) Let $\alpha \in [\mathfrak{c}, \mathfrak{c}^+]^V$ and let us verify $j(\omega_1^V) > \alpha$.
- (5) There is $A \subseteq \mathbb{R}^V$ such that A codes a well-ordering of $\mathbb{R}^V$ of order-type α .
- (6) We have $A = j[A] = j[\mathbb{R}] \cap j(A) \in \mathfrak{M}$. Then model $\mathfrak{M}$ computes α as the collapse of the well-ordering codes by A .
- (7) We want to show that A is countable in $\mathfrak{M}$. Since $A \subseteq \mathbb{R}^V$, it suffices to show $|\mathbb{R}^V|^{\mathfrak{M}} \leq \omega$.
- (8) We reason as follows:

$$|\mathbb{R}^V|^{\mathfrak{M}} = |[\text{id}]|^{\mathfrak{M}} = |([\sigma] : \sigma \in [\mathbb{R}]^\omega)| \leq |(\omega : \sigma \in [\mathbb{R}]^\omega)| = j(\omega) = \omega.$$

- (9) Since $|A|^{\mathfrak{M}} = \omega$, we conclude $|\alpha|^{\mathfrak{M}} = \omega$ and consequently $\alpha < \omega_1^{\mathfrak{M}} = j(\omega_1^V)$.

$\square$

[1.3.13] **Definition.** Normal precipitous ideal I on $[\mathbb{R}]^\omega$ is *strong* if

$$\mathbb{P}_I \Vdash j_{\dot{G}}(\omega_1^V) = \mathfrak{c}^{+V}.$$

[1.3.14] **Lemma.** Let $f : \mathbb{R} \rightarrow \alpha^\omega$ be a surjection and let $\pi(\sigma) := f[\sigma]$ for $\sigma \in [\mathbb{R}]^\omega$. Then

$$V^{\mathbb{P}_I} \models j_{\dot{G}}[\alpha^\omega] = [\pi]_{\dot{G}}.$$

Proof.

- (1) Let $g : [\mathbb{R}]^\omega \rightarrow V$, let $G \rightsquigarrow V^{\mathbb{P}_I}$, and let $j := j_G$. We want to show

$$V[G] \models [g] \in j[\alpha^\omega] \iff [g] \in [\pi].$$

- (2) Assume first $[g] \in j[\alpha^\omega]$. Let us show $[g] \in [\pi]$.

(2.1) There exists $x \in \mathbb{R}^V$ such that $[g] = j(f(x))$.

(2.2) $\{\sigma : g(\sigma) = f(x)\} \in G$

(2.3) $\{\sigma : x \in \sigma\} \in G$

(2.4) If $x \in \sigma$ and $g(\sigma) = f(x)$, then $g(\sigma) \in f[\sigma] = \pi(\sigma)$.

(2.5) Hence, $\{\sigma : g(\sigma) \in \pi(\sigma)\} \in G$.

(2.6) Hence, $[g] \in [\pi]$.

(3) Assume now $[g] \in [\pi]$. Let us verify $[g] \in j_G[\alpha^\omega]$.

(3.1) $A := \{\sigma : g(\sigma) \in f[\sigma]\} \in G$

(3.2) In V , there exists $h : A \rightarrow V$ such that for every $\sigma \in A$, we have $h(\sigma) \in \sigma$ and $g(\sigma) = f(h(\sigma))$.

(3.3) Hence, $[g] = j(f)([h])$ and $[h] \in [\text{id}]$.

(3.4) By normality, $[h] = j(x)$ for some $x \in \mathbb{R}^V$.

(3.5) Hence, $[g] = j(f)(j(x)) = j(f(x)) \in j[\alpha^\omega]$.

□

[1.3.15] **Lemma.** Let $\alpha < \Theta = \mathfrak{c}^+$. Then $V^{\mathbb{P}_I} \models j_{\dot{G}}[\alpha^\omega] \in \text{Ult}(V, \dot{G})$.

□

[1.3.16] **Definition.** Ideal I is *pseudo-homogeneous* if it is precipitous and for every $\alpha < \text{Ord}$, every $s \in \text{Ord}^\omega$, every $\lambda < \Theta$, and every formula $\phi(x, y, z)$, we have that the truth value of the statement

$$\text{Ult}(V, \dot{G}) \models \phi[\alpha, j_{\dot{G}}(s), j_{\dot{G}}[\lambda^\omega]]$$

is decided by the weakest condition.

[1.3.17] **Definition.** An ideal I on X is κ -dense if there exists a dense set $D \subseteq P(X)/I$ of cardinality $\leq \kappa$. An ideal on ω_1 is *dense* if it is ω_1 -dense.

[1.3.18] **Definition.** An ideal I on X is *predictable* if it is precipitous and if for every $\xi \in \text{Ord}$, there exists $\eta \in \text{Ord}$ such that $\mathbb{P}_I \Vdash j_{\dot{G}}(\xi) = \eta$.

[1.3.19] **Lemma.** Let J be a normal ideal on $[\omega_1]^\omega$. Then $I := J \cap \mathcal{P}(\omega_1)$ is a normal ideal on ω_1 .

Proof.

(1) It is clear that I is an ideal.

(2) For every $\alpha < \omega_1$, we have

$$\alpha \subseteq \{\sigma \in [\omega_1]^\omega : \alpha \notin \sigma\} \in J,$$

which implies $\alpha \in I$.

(3) It remains to verify that I is closed for diagonal unions. However, this is obvious since diagonal unions in the sense of ω_1 are the same as diagonal unions in the sense of $[\omega_1]^\omega$.

□

[1.3.20] **Lemma.** Let J be a normal ideal on $[\omega_1]^\omega$. Then J is countably closed.

Proof.

(1) Let F be the filter dual to J .

(2) **Claim.** $X := \{\sigma : \omega \subseteq \sigma\} \in F$

Proof.

(2.1) Let $S_\alpha := \{\sigma : \alpha + 1 \in \sigma\} \in F$ for $\alpha < \omega_1$. Let $T := \{\sigma : 0 \in \sigma\} \in F$.

(2.2) Since F is normal, we have

$$U := \bigtriangleup_{\alpha < \omega_1} S_\alpha \cap T \in F.$$

(2.3) If $\sigma \in U$, then σ contains 0 and is closed for the successor. In particular, $\omega \subseteq \sigma$.

□

(3) Let $S_n \in J$ for $n < \omega$. We want to show $\bigcup_{n < \omega} S_n \in J$.

(4) Let $S_\alpha := \emptyset$ for $\alpha \in [\omega, \omega_1)$. We have by normality that

$$\bigvee_{\alpha < \omega_1} S_\alpha \in J.$$

(5) We also have

$$\bigvee_{\alpha < \omega_1} S_\alpha \cap X = \bigcup_{n < \omega} S_n \cap X.$$

Since $X \in F$, we conclude $\bigcup_{n < \omega} S_n \in J$.

□

[1.3.21] **Lemma.** Let J be a normal ideal on $[\omega_1]^\omega$, let $I := J \cap \mathcal{P}(\omega_1)$, and let $S \in J^+$. Then $S \cap \omega_1 \in I^+$.

Proof.

- (1) It suffices to show that $S \cap \omega_1 \notin J$.
- (2) For this in turn it suffices to show that $\omega_1 \in F$, where F is the filter dual to J .
- (3) Assume towards contradiction that $\omega_1 \notin F$. This means that $T := [\omega_1]^\omega - \omega_1 \notin J$.
- (4) Let $f : T \rightarrow \omega_1$ be defined by mapping $\sigma \in T$ to the smallest $\alpha \in \sigma$ with $\alpha \notin \sigma$.
- (5) Since J is normal, there exist $U \in J^+$ and $\alpha < \omega_1$ such that $U \subseteq T$ and $f(\sigma) = \alpha$ for every $\sigma \in U$.
- (6) For every $\sigma \in U$, there exists $\beta < \alpha$ such that $\beta \notin \sigma$. Since J is countably closed, there exists $U' \in J^+$ and $\beta < \alpha$ such that $U' \subseteq U$ and $\beta \notin \sigma$ for every $\sigma \in U'$.
- (7) This means $U' \subseteq \{\sigma : \beta \notin \sigma\} \in J$, which is a contradiction.

□

[1.3.22] **Lemma.** Let J be a normal ω_1 -dense ideal on $[\omega_1]^\omega$ and let $I := J \cap \mathcal{P}(\omega_1)$. Then I is a normal dense ideal on ω_1 .

Proof.

- (1) Let $(T_\alpha : \alpha < \omega_1) \in (J^+)^{\omega_1}$ be such that for every $T \in J^+$, there exists $\alpha < \omega_1$ satisfying $T_\alpha - T \in J$.
- (2) Let $S_\alpha := T_\alpha \cap \omega_1 \in I^+$ for $\alpha < \omega_1$. We want to show that for every $S \in I^+$, there exists $\alpha < \omega_1$ such that $S_\alpha - S \in I$.
- (3) Let $S \in I^+$ be arbitrary. We have $S \notin J$, i.e. $S \in J^+$.
- (4) Hence, there exists $\alpha < \omega_1$ such that $T_\alpha - S \in J$.
- (5) We have $S_\alpha - S = (T_\alpha - S) \cap \omega_1 \subseteq T_\alpha - S \in J$ and consequently $S_\alpha - S \in I$.

□

[1.3.23] **Lemma.** Let J be a normal predictable ideal on $[\omega_1]^\omega$ and let $I := J \cap \mathcal{P}(\omega_1)$. Then I is a normal predictable ideal on ω_1 .

Proof.

- (1) Let $g \rightsquigarrow V^{\mathbb{P}_I}$ be arbitrary and let us show that $\text{Ult}(V, g)$ is well-founded and $j_g \restriction \text{Ord}$ does not depend on g .
- (2) Let $h := \{S \subseteq [\omega_1]^\omega : S \cap \omega_1 \in g\}$. It is clear that h is a filter on $\mathbb{P}_J$.
- (3) If D is a dense subset of $\mathbb{P}_J$, then $\{S \cap \omega_1 : S \in D\}$ is a dense subset of $\mathbb{P}_I$. This implies that there exists $S \in S$ such that $S \cap \omega_1 \in h$ and consequently $S \in D \cap h$.
- (4) Hence, $h \rightsquigarrow V^{\mathbb{P}_J}$, model $\text{Ult}(V, h)$ is well-founded, and $j_h \restriction \text{Ord}$ does not depend on h .

(5) We will be done with the argument if we show that

$$k : \text{Ult}(V, h) \rightarrow \text{Ult}(V, g) : [f]_h \mapsto [f \restriction \omega_1]_g$$

is a correctly defined elementary embedding, that diagram

$$\begin{array}{ccc} V & & \\ \downarrow j_h & \searrow j_g & \\ \text{Ult}(V, h) & \xrightarrow{k} & \text{Ult}(V, g) \end{array}$$

commutes, and that k is the identity.

(6) To see that k is correctly defined and elementary, let $\phi(x)$ be a formula, let $f \in ([\omega_1]^\omega \rightarrow V)^V$, and suppose $\text{Ult}(V, h) \models \phi([f]_h)$. We want to show that $\text{Ult}(V, g) \models \phi([f \restriction \omega_1^V]_g)$.

(7) We have

$$\{\sigma \in [\omega_1]^\omega : \phi(f(\sigma))\}^V \in h.$$

By definition of h , this means that

$$\{\alpha < \omega : \phi(f(\alpha))\}^V \in g,$$

so $\text{Ult}(V, g) \models \phi([f \restriction \omega_1^V]_g)$.

(8) It is obvious that the diagram commutes, so it remains to verify that k is the identity. It suffices to verify that it is surjective.

(9) Let $u \in (\omega_1 \rightarrow V)^V$ be arbitrary. We want to find $v \in ([\omega_1]^\omega \rightarrow V)^V$ such that

$$\{\alpha < \omega_1^V : u(\alpha) = v(\alpha)\} \in g.$$

(10) Let

$$v(\sigma) := \begin{cases} u(\alpha), & \sigma = \alpha < \omega_1 \\ 0, & \text{otherwise.} \end{cases}$$

We clearly have $\{\alpha < \omega_1^V : u(\alpha) = v(\alpha)\} = \omega_1 \in g$.

□

1.4 Prewellorderings and cardinal Θ

[1.4.1] **Definition.** A *prewellordering* on X is a binary relation $\leq$ such that:

- a. $(\forall x \in X)(x \leq x)$
- b. $(\forall x, y, z \in X)(x \leq y \wedge y \leq z \implies x \leq z)$
- c. $(\forall x, y \in X)(x \leq y \vee y \leq x)$
- d. the relation $<$ defined by

$$x < y : \iff x \leq y \wedge y \not\leq x$$

is well-founded.

Corresponding equivalence relation $\sim$ is defined as follows

$$x \sim y : \iff x \leq y \wedge y \leq x.$$

[1.4.2] **Remark.**

- a. $\sim$ is indeed an equivalence relation.
- b. $<$ is irreflexive, transitive, well-founded, and

$$(\forall x, y \in X)(x < y \vee x \sim y \vee y < x).$$

- c. $x < y \iff x \leq y \wedge x \not\sim y$

- d. $(X, \leq)/\sim$ is a well-ordering.
e. A binary relation $\leq$ on X is a prewellordering if and only if there exists a mapping $\phi : X \rightarrow \text{Ord}$ such that

$$(\forall x, y \in X)(x \leq y \iff \phi(x) \leq \phi(y)).$$

- f. The length of a prewellordering $\leq$ is defined to be the order-type of $(X, \leq)/\sim$.
g. The length of a prewellordering $\leq$ is the minimal ordinal α such that

$$(\exists \phi : X \rightarrow \alpha)(\forall x, y \in X)(x \leq y \iff \phi(x) \leq \phi(y)).$$

[1.4.3] **Definition.**

- a. $\Theta_X := \sup\{\text{length}(\leq) + 1 : \leq \subseteq X \times X \text{ is a prewellordering of } X\}$
b. $\Theta := \Theta_{\mathbb{R}}$

[1.4.4] **Proposition.**

- a. Θ_X is the smallest ordinal $\alpha > 0$ such that there is no surjection $\phi : X \rightarrow \alpha$.
b. $\Theta = \sup\{\Theta^{L(\mathbb{R}; A)} : A \subseteq \mathbb{R}\}$

□

[1.4.5] **Declaration.** We assume for the rest of the subsection that $V \models \text{AD} + \text{DC}_{\mathbb{R}}$.

[1.4.6] **Definition.**

- a. $A \subseteq \mathbb{R}$ is said to be *Wadge reducible* to $B \subseteq \mathbb{R}$ if there exists continuous $f : \mathbb{R} \rightarrow \mathbb{R}$ such that $A = f^{-1}[B]$. We write this as $A \leq_W B$.
b. *Modified Wadge reducibility* $\leq_W^*$ is defined as

$$A \leq_W^* B : \iff A \leq_W B \vee A \leq_W B^{\mathcal{C}}.$$

[1.4.7] **Lemma.**

- a. $\leq_W$ is reflexive and transitive.
b. $A \leq_W B$ if and only if $A^{\mathcal{C}} \leq_W B^{\mathcal{C}}$.
c. $\leq_W^*$ is reflexive and transitive.

[1.4.8] **Lemma.**

- a. For every $A, B \subseteq \mathbb{R}$, either $A \leq_W B$ or $B \leq_W A^{\mathcal{C}}$.
b. For every $A \subseteq \mathbb{R}$, if $A \leq_W A^{\mathcal{C}}$, then $A \sim_W A^{\mathcal{C}}$.
c. For every $A, B \subseteq \mathbb{R}$, either $(A <_W B, B^{\mathcal{C}})$ or $(A \sim_W B \vee A \sim_W B^{\mathcal{C}})$ or $(B, B^{\mathcal{C}} <_W A)$.
d. For every $A, B \subseteq \mathbb{R}$, either $A \leq_W^* B$ or $B \leq_W^* A$.
e. For every $A, B \subseteq \mathbb{R}$:
i. if $A <_W B, B^{\mathcal{C}}$, then $A <_W^* B \sim_W^* B^{\mathcal{C}}$;
ii. if $A \sim_W B$ or $A \sim_W B^{\mathcal{C}}$, then $A \sim_W^* B \sim_W^* B^{\mathcal{C}}$;
iii. if $B, B^{\mathcal{C}} <_W A$, then $B \sim_W^* B^{\mathcal{C}} <_W^* A$.

Proof.

- (1) We show **a**; **b** is obvious, while **c**, **d**, **e** follow from **a** and **b**.
(2) Consider the following game:

I	x_0	x_1	$\cdots$
II	y_0	y_1	$\cdots$

where $x_i, y_i < \omega$; player I wins if

$$(x_i : i < \omega) \in A \iff (y_i : i < \omega) \in B^{\mathcal{C}}.$$

- (3) If player I wins this game, then their winning strategy determines continuous mapping $f : \mathbb{R} \rightarrow \mathbb{R}$ satisfying $B^{\mathbb{C}} = f^{-1}[A]$. Thus, $B^{\mathbb{C}} \leq_W A$ and $B \leq_W A^{\mathbb{C}}$.
- (4) If player II wins this game, then their winning strategy determines continuous mapping $f : \mathbb{R} \rightarrow \mathbb{R}$ satisfying $A = f^{-1}[B]$. Thus, $A \leq_W B$.

□

[1.4.9] **Proposition.** Relation $<_W$ is well-founded.

[1.4.10] **Corollary.** Relation $\leq_W^*$ is a prewellordering on $\mathcal{P}(\mathbb{R})$.

□

[1.4.11] **Definition.** The *Wadge rank* of a set $A \subseteq \mathbb{R}$ is its rank w.r.t. the relation $<_W$.

[1.4.12] **Remark.** Equivalently, it is its rank w.r.t. $<_W^*$, or in other words, the order type of

$$\{B / \sim_W^* : B <_W^* A\}$$

order by the quotient $<_W^* / \sim_W^*$.

[1.4.13] **Theorem.** Ordinal Θ is the rank of relation $<_W$, i.e.

$$\Theta = \sup\{\text{rank}_W(A) + 1 : A \subseteq \mathbb{R}\}.$$

[1.4.14] **Remark.** Equivalently, Θ is the order type of

$$(\mathcal{P}(\mathbb{R}), \leq_W^*) / \sim_W^*.$$

[1.4.15] **Corollary.** Let M be a transitive model for ZF such that $\mathbb{R} \subseteq M$. Then $\Theta^M \leq \Theta$. If $\mathcal{P}(\mathbb{R}) \not\subseteq M$, then $\Theta^M < \Theta$.

Proof.

(1) Since continuous functions $f : \mathbb{R} \rightarrow \mathbb{R}$ are coded by reals, we have:

- a. $(\forall A \in \mathcal{P}(\mathbb{R}) \cap M)(\forall B \in \mathcal{P}(\mathbb{R})) (B \leq_W A \implies B \in M)$
- b. $(\forall A, B \in \mathcal{P}(\mathbb{R}) \cap M) (A \leq_W^M B \iff A \leq_W B)$.

(2) Since $\mathbb{R} \subseteq M$, we have $M \models \text{AD} + \text{DC}_{\mathbb{R}}$.

(3) Thus,

$$\Theta^M = \text{rank}(\mathcal{P}(\mathbb{R}), \leq_W)^M \leq \text{rank}(\mathcal{P}(\mathbb{R}), \leq_W) = \Theta.$$

(4) If there is $A \in \mathcal{P}(\mathbb{R}) - M$, then for every $B \in \mathcal{P}(\mathbb{R}) \cap M$, we have $B <_W A$.

(5) We conclude

$$\Theta^M = \sup\{\text{rank}_W(B) + 1 : B \in \mathcal{P}(\mathbb{R}) \cap M\} \leq \text{rank}_W(A) < \Theta.$$

□

[1.4.16] **Definition.** Let $\alpha < \Theta$. Then we define $\mathcal{P}_\alpha(\mathbb{R}) := \{A \subseteq \mathbb{R} : \text{the Wadge rank of } A \text{ is } < \alpha\}$.

1.5 Some consequences of $\text{AD}_{\mathbb{R}}$

[1.5.1] **Summary.** We work in this subsection under the assumption that $\text{AD}_{\mathbb{R}}$ holds.

[1.5.2] **Definition.** Set $A \subseteq \mathbb{R}^2$ is *uniformizable* if there exists $f : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$(\forall x \in \mathbb{R})(\exists y)((x, y) \in A \implies (x, f(x)) \in A).$$

[1.5.3] **Theorem.** Every set of reals is uniformizable.

Proof. See Theorem 0.1.2. of [Larson, 2019].

□

[1.5.4] **Proposition.** Suppose that $\text{cof}(\Theta) > \omega$. Then $\text{DC}_{\mathcal{P}(\mathbb{R})}$ holds.

Proof.

- (1) Let T be a tree on $\mathcal{P}(\mathbb{R})$ with no leaves and let us show that T has a branch.
- (2) Since $\text{cof}(\Theta) > \omega$, there exists some $\alpha < \Theta$ such that $S := T \cap \mathcal{P}_\alpha(\mathbb{R})^{<\omega}$ is a tree on $\mathcal{P}_\alpha(\mathbb{R})$ without leaves. It suffices to show that S has a branch.
- (3) Let $A \subseteq \mathbb{R}$ be of Wadge-rank α and let U be the following tree on $\mathbb{R}$: tuple $(x_0, \dots, x_{n-1})$ belongs to U if and only if every x_i codes a continuous function $f_i : \mathbb{R} \rightarrow \mathbb{R}$ and $(f_i^{-1}[A] : i < n) \in S$.
- (4) Let us fix a canonical coding $\langle \cdot \rangle : \mathbb{R}^{<\omega} \rightarrow \mathbb{R}$ and let us define $R \subseteq \mathbb{R}^2$ as follows: $(x, y) \in R$ if and only if $x = \langle s \rangle$ for some $s \in \mathbb{R}^{<\omega}$ and $s \frown y \in S$.
- (5) By [1.5.3], there exists $f : \mathbb{R} \rightarrow \mathbb{R}$ such that for every $x \in \mathbb{R}$

$$\exists y((x, y) \in R) \implies (x, f(x)) \in R.$$

- (6) Let $x \in \mathbb{R}^\omega$ be defined recursively as follows:

$$x(n) := f(\langle x \upharpoonright n \rangle) \quad (n < \omega).$$

Clearly $x \in [U]$.

- (7) For $n < \omega$, let $f_n : \mathbb{R} \rightarrow \mathbb{R}$ be the continuous function coded by x_n . We have

$$(f_n^{-1}[A] : n < \omega) \in [S].$$

□

[1.5.5] **Corollary.** Assume that $V = L(\mathcal{P}(\mathbb{R}))$ and $\text{cof}(\Theta) > \omega$. Then DC holds.

Proof. This follows from the previous proposition and condition [1.1.6].

□

[1.5.6] **Proposition.** Assume that DC holds. Then every set of reals is Suslin.

Proof. See Theorem 0.1.2. of [Larson, 2019].

□

[1.5.7] **Goal.** We wish now to prove that the club filter on $[\mathbb{R}]^\omega$ is an ultrafilter. We will first define and study a filter denoted by μ and we will later conclude that μ is exactly the club filter.

[1.5.8] **Definition.** For $A \subseteq [\mathbb{R}]^\omega$, game $G^*(A)$ is played as follows:

I	x_0	x_2	$\dots$
II	x_1	x_4	$\dots$

where $x_i \in \mathbb{R}$. Player I wins $\{x_i : i < \omega\} \in A$; otherwise, II wins.

[1.5.9] **Exercise.** Game $G^*(A)$ is determined for every $A \subseteq [\mathbb{R}]^\omega$.

[1.5.10] **Definition.** $\mu \subseteq \mathcal{P}([\mathbb{R}]^\omega)$ is defined as follows: $A \in \mu$ if and only if player I wins $G^*(A)$.

[1.5.11] **Proposition.** Object μ is a countably complete ultrafilter on $\mathcal{P}([\mathbb{R}]^\omega)$.

Proof.

- (1) Clearly $[\mathbb{R}]^\omega \in \mu$ and $\emptyset \notin \mu$.
- (2) Let $A \subseteq [\mathbb{R}]^\omega$ be arbitrary and let us verify $A \in \mu$ or $A^c \in \mu$. Assume otherwise.
- (3) Then player II wins $G^*(A)$ by some σ and $G^*(A^c)$ by some σ' .
- (4) We produce plays

I	x_0	x_1	$\cdots$
II	y_0	y_1	$\cdots$
I	z_0	z_1	$\cdots$
II	w_0	w_1	$\cdots$

as follows:

- a. x_0 is arbitrary;
- b. $y_n := \sigma(x_0, \dots, x_n)$
- c. $z_0 := x_0$
- d. $w_n := \sigma'(z_0, \dots, z_n)$
- e. $x_{n+1} := w_n$
- f. $z_{n+1} := y_n$.

- (5) Since the first board is played according to σ , we have

$$\{x_0\} \cup \{y_i : i < \omega\} \cup \{w_i : i < \omega\} = \{x_i : i < \omega\} \cup \{y_i : i < \omega\} \in A^{\mathbb{G}}.$$

Since the second board is played according to σ' , we have

$$\{x_0\} \cup \{y_i : i < \omega\} \cup \{w_i : i < \omega\} = \{z_i : i < \omega\} \cup \{w_i : i < \omega\} \in A.$$

This is obviously a contradiction.

- (6) Let us verify countable completeness. Assume that it fails and let $(A_n : n < \omega)$ be such that player I wins $G^*(A_n)$ via σ_n , while player II wins $G^*(\bigcap_{n < \omega} A_n)$ via τ .
- (7) We produce plays

I	x_0^i	x_1^i	$\cdots$
II	y_0^i	y_1^i	$\cdots$

of $G^*(A_i)$ played according to σ_i ($i < \omega$) and also a play

I	z_0	z_1	$\cdots$
II	w_0	w_1	$\cdots$

played according to τ .

- (8) x_j^i 's and w_j 's are determined by the strategies, while y_j^i 's and z_j 's can be chosen freely.
- (9) Use bookkeeping to ensure that after ω moves every board has the same content seen as a set of reals. Let a be that set of reals.
- (10) Since a is produced when playing $G^*(A_i)$ according to σ_i , we have $a \in A_i$ for every i .
- (11) Since a is produced when playing $G^*(\bigcap_i A_i)$ according to τ , we have $a \in \bigcap_i A_i$ for every i . This is a contradiction.

□

[1.5.12] **Proposition.** Ultrafilter μ is normal.

Proof.

- (1) To verify fineness, observe that in order to win $G^*(\{a \in [\mathbb{R}]^\omega : x \in a\})$, player I simply needs to play x in their first move.
- (2) To verify normality, let $F : \mathbb{R} \rightarrow \mu^{\mathbb{R}}$ be arbitrary and assume towards contradiction that $(\Delta F)^{\mathbb{G}} \in \mu$.
- (3) By $\text{AC}_{\mathbb{R}}$, we can choose winning strategies $(\sigma_x : x \in \mathbb{R})$ for player I in games $G^*(F(x))$ ($x \in \mathbb{R}$). Let τ be a winning strategy for player II in $G^*(\Delta F)$.
- (4) We now proceed towards contradiction in the fashion similar to that of the proof of countable completeness. The difference is that countably many relevant boards are not known in the beginning, but rather we start with just a board for $G^*(\Delta F)$ and then as some real x is played on some of the existing boards, we add a new board for $G^*(F(x))$.

- (5) Using a bookkeeping device, we ensure that all the boards in the end have the same content $a \in [\mathbb{R}]^\omega$.
- (6) For $x \in a$, the board/one of the boards for $G^*(F(x))$ is played according to σ_x , ensuring $a \in F(x)$. Thus, $a \in \bigcap_{x \in a} F(x)$ and consequently $a \in \Delta F$.
- (7) This is a contradiction since the board $G^*(\Delta F)$ is played according to τ , ensuring $a \notin \Delta F$.

□

[1.5.13] **Theorem.** The club filter on $[\mathbb{R}]^\omega$ is equal to μ . In particular, the club filter on $[\mathbb{R}]^\omega$ is a normal ultrafilter.

Proof.

- (1) Let $C \subseteq [\mathbb{R}]^\omega$ be club and let us show $C \in \mu$.
- (2) There is some $F : \mathbb{R}^{<\omega} \rightarrow \mathbb{R}$ such that

$$\{a \in [X]^\omega : F[a^{<\omega}] \subseteq a\} \subseteq C.$$

- (3) A winning strategy for player I in $G^*(A)$ is obtained as follows: use bookkeeping to ensure that the play is closed for F .
- (4) Let now $A \in \mu$ and let us verify that A is a club.
- (5) Let $\sigma : \mathbb{R}^{<\omega} \rightarrow \mathbb{R}$ be a winning strategy for player I in $G^*(A)$. We are done if we show that for every $a \in [\mathbb{R}]^\omega$ with $\sigma[a^{<\omega}] \subseteq a$, we have $a \in A$.
- (6) Let player I play according to σ and let player II play some enumeration of a . This is a play of $G^*(A)$ according to σ .
- (7) This play produces exactly set a , and since it is won by player I, we have $a \in A$.

□

2 Ultrapowers via the Club Ultrafilter

[2.0.1] **Declaration.** We assume $V \models \text{AD}_{\mathbb{R}} + \text{DC}$. The club ultrafilter is denoted by μ .

[2.0.2] **Remark.** We do not have Łoś's theorem for $\text{Ult}(V, \mu)$ in absence of choice. In particular, $j := j_\mu : V \rightarrow \text{Ult}(V, \mu)$ might not be elementary.

[2.0.3] **Proposition.** If M is an inner model, $\phi(x_0, \dots, x_{n-1})$ a formula, $f_0, \dots, f_{n-1} : [\mathbb{R}]^\omega \rightarrow M$ functions, and

$$j_*M := \{f : [\mathbb{R}]^\omega \rightarrow M\} / \mu,$$

then

$$j_*M \models \phi[[f_0], \dots, [f_{n-1}]] \iff \{a \in [\mathbb{R}]^\omega : M \models \phi[f_0(a), \dots, f_{n-1}(a)]\} \in \mu.$$

□

[2.0.4] **Proposition.** If M is an inner model, then j_*M is well-founded and

$$j \restriction M : M \prec j_*M.$$

Proof.

- (1) Elementarity follows from [2.0.3].
- (2) Let us first verify well-foundedness. Assume otherwise and choose a sequence $f_i : [\mathbb{R}]^\omega \rightarrow \text{Ord}$ ($i < \omega$) such that $[f_{i+1}]_\mu < [f_i]_\mu$ for every i . This can be done by DC.
- (3) We have $A_i := \{a \in [\mathbb{R}]^\omega : f_{i+1}(a) < f_i(a)\} \in \mu$ for every i .
- (4) By countable completeness, we have

$$\bigcap_{i < \omega} A_i \neq \emptyset,$$

which would yield a descending ω -sequence of ordinals. Contradiction.

□

[2.0.5] **Corollary.** $j \restriction \text{Ord} : \text{Ord} \rightarrow \text{Ord}$

□

[2.0.6] **Proposition.** Suppose that for every $\sigma \in [\mathbb{R}]^\omega$, we have $A_\sigma \subseteq L(\sigma)$. Let $\phi(x_0, \dots, x_{n-1})$ be a formula and let f_i be functions on $[\mathbb{R}]^\omega$ such that $f_i(\sigma) \in L(\sigma; A_\sigma)$ for every $\sigma \in [\mathbb{R}]^\omega$. Then

$$\prod_{\sigma \in [\mathbb{R}]^\omega} L(\sigma; A_\sigma) / \mu \models \phi[[f_0], \dots, [f_{n-1}]] \iff \{\sigma \in [\mathbb{R}]^\omega : L(\sigma; A_\sigma) \models \phi[f_0(\sigma), \dots, f_{n-1}(\sigma)]\} \in \mu.$$

Proof.

- (1) This is proved inductively on ϕ . Only the $\exists$ step is non-trivial, and only ($\Leftarrow$) implication of that step.
- (2) Suppose

$$B := \{\sigma \in [\mathbb{R}]^\omega : L(\sigma; A_\sigma) \models \exists y \phi(f(\sigma), y)\} \in \mu.$$

We want to find $g \in \prod \{L(\sigma; A_\sigma) : \sigma \in [\mathbb{R}]^\omega\}$ such that

$$\prod_{\sigma \in [\mathbb{R}]^\omega} L(\sigma; A_\sigma) / \mu \models \phi[[f], [g]].$$

- (3) We may assume w.l.o.g. that σ is closed for the coding of finite tuples of σ .
- (4) For every $\sigma \in B$, there exist a formula ψ , an ordinal $\alpha \in \text{Ord}$, and a real $x \in \sigma$ such that ψ defines with parameters α, x, A_σ over $L(\sigma; A_\sigma)$ a set y satisfying

$$L(\sigma; A_\sigma) \models \phi[f(\sigma), y].$$

(5) For $\sigma \in B$, let $(\psi_\sigma, \alpha_\sigma)$ be the lexicographically minimal pair such that there exists $x \in \sigma$ satisfying:

- a. ψ_σ defines with parameters $\alpha_\sigma, x, A_\sigma$ over $L(\sigma; A_\sigma)$ some set $y_{\sigma, x}$;
- b. $L(\sigma, A_\sigma) \models \phi[f(\sigma), y_{\sigma, x}]$.

(6) Let F be a function on $[\mathbb{R}]^\omega$ defined by

$$F(\sigma) := \{x \in \sigma : (y_{\sigma, x} \text{ is defined}) \wedge L(\sigma; A_\sigma) \models \phi[f(\sigma), y_{\sigma, x}]\}.$$

We have $\emptyset \subset F(\sigma) \subseteq \sigma$ for every $\sigma \in B \in \mu$.

(7) By normality, there exist $x_0 \in \mathbb{R}$ and $B_0 \in \mu$ such that $B_0 \in \mu$ and $x_0 \in \sigma$ for every $\sigma \in B_0$.

(8) Let

$$g \in \prod_{\sigma \in [\mathbb{R}]^\omega} L(\sigma; A_\sigma)$$

be defined as follows:

$$g(\sigma) := \begin{cases} y_{\sigma, x_0}, & \sigma \in B_0 \\ 0, & \text{otherwise.} \end{cases}$$

(9) We have

$$\mu \ni B_0 \subseteq \{\sigma \in [\mathbb{R}]^\omega : L(\sigma; A_\sigma) \models [f(\sigma), [g]]\},$$

so by the IH, we conclude

$$\prod_{\sigma \in [\mathbb{R}]^\omega} L(\sigma; A_\sigma) / \mu \models \phi[[f], [g]].$$

□

[2.0.7] **Corollary.** Suppose that for every $\sigma \in [\mathbb{R}]^\omega$, we have $A_\sigma \subseteq L(\sigma)$. Then $A := [\sigma \mapsto A_\sigma] \subseteq L(\mathbb{R})$ and

$$\prod_{\sigma \in [\mathbb{R}]^\omega} L(\sigma; A_\sigma) / \mu = L(\mathbb{R}; A).$$

□

[2.0.8] **Corollary.** $A^\sharp$ exists for every $A \subseteq \mathbb{R}$.

Proof.

- (1) For $\sigma \in [\mathbb{R}]^\omega$ and $A \subseteq \sigma$, pair set $\sigma \cup \{A\}$ can be coded by a single real.
- (2) Since every real has the sharp, we conclude that every such pair $(\sigma; A)$ has the sharp.
- (3) Let now $A \subseteq \mathbb{R}$ be arbitrary and let us show that $(\mathbb{R}; A)^\sharp$ exists.
- (4) For every $\sigma \in [\mathbb{R}]^\omega$ and for $A_\sigma := A \cap \sigma$ such that $\sigma = \mathbb{R}^{L(\sigma; A_\sigma)}$, there exists $(\sigma; A_\sigma)^\sharp \subseteq \sigma$.
- (5) Since $\mathbb{R} = \mathbb{R}^{L(\mathbb{R}; A)}$, we have

$$\forall^\mu \sigma (\sigma = \mathbb{R}^{L(\sigma; A_\sigma)}).$$

Thus, we can define

$$S := [(\sigma \mapsto (\sigma; A_\sigma)^\sharp) : \sigma \in [\mathbb{R}]^\omega, \sigma = \mathbb{R}^{L(\sigma; A_\sigma)}] \subseteq \mathbb{R}.$$

(6) We have $L(\mathbb{R}; S) = \prod_\sigma L(\sigma; (\sigma; A_\sigma)^\sharp) / \mu$, so by Łoś's theorem,

$$L(\mathbb{R}; A) \models S = ([\sigma \mapsto \sigma]; [\sigma \mapsto A_\sigma])^\sharp = (\mathbb{R}; A)^\sharp.$$

□

[2.0.9] **Lemma.** Let $g \rightsquigarrow V^{\text{Col}(\omega, \mathbb{R})}$. Then $\omega_1^{V[g]} \leq \Theta^V$.

□

[2.0.10] **Proposition.** $j(\omega_1) = \Theta$

Proof.

- (1) We first verify $\Theta \leq j(\omega_1)$. Let $A \subseteq \mathbb{R}$ be arbitrary and let us verify $\Theta^{L(\mathbb{R};A)} \leq j(\omega_1)$.
- (2) Let $A_\sigma := A \cap \sigma$ for $\sigma \in [\mathbb{R}]^\omega$. We have

$$\prod_{\sigma \in [\mathbb{R}]^\omega} L(\sigma; A_\sigma) / \mu = L([\sigma \mapsto \sigma]; [\sigma \mapsto A_\sigma]) = L(\mathbb{R}; A).$$

- (3) Thus,

$$\Theta^{L(\mathbb{R};A)} = [\sigma \mapsto \Theta^{L(\sigma;A_\sigma)}].$$

- (4) Since a pair $(\sigma; A_\sigma)$ can be coded by a single real x and since $\Theta^{L(\sigma;A_\sigma)}$ is definable in $L[x]$ from x alone, we have that $\Theta^{L(\sigma;A_\sigma)}$ is $<$ the first indiscernible for $L[x]$. In particular,

$$\Theta^{L(\sigma;A_\sigma)} < \omega_1$$

- (5) We thus conclude

$$\Theta^{L(\mathbb{R};A)} < j(\omega_1)$$

and since $A \subseteq \mathbb{R}$ was arbitrary, $\Theta \leq j(\omega_1)$.

- (6) To see that $j(\omega_1) \leq \Theta$, let $f : [\mathbb{R}]^\omega \rightarrow \omega_1$ be arbitrary and let us verify that $[f] < \Theta$.
- (7) Let A be the set of all $(x, y) \in \mathbb{R}^2$ such that x codes $\sigma_x \in [\mathbb{R}]^\omega$ and y codes $\alpha_y < \omega_1$ satisfying $f(\sigma_x) = \alpha_y$.
- (8) Let T be a tree on $\omega \times \omega \times \gamma$ such that $p[T] = A$.
- (9) For $\sigma \in [\mathbb{R}]^\omega$, let $W_\sigma := L(\sigma; T)$, and let

$$W := \prod_{\sigma \in [\mathbb{R}]^\omega} W_\sigma = L(\mathbb{R}; j(T)).$$

- (10) **Claim.** $\{\sigma \in [\mathbb{R}]^\omega : f(\sigma) < \Theta^{W_\sigma}\} \in \mu$

Proof.

- (10.1) $B := \{\sigma \in [\mathbb{R}]^\omega : \mathbb{R}^{W_\sigma} = \sigma\} \in \mu$
- (10.2) Let $\sigma \in B$ and let $x \in \mathbb{R}$ be a real coding it.
- (10.3) We have $W_\sigma \subseteq L[x, T] \models \text{ZFC}$.
- (10.4) Since ω_1^V is inaccessible in $L[x, T]$, we have that $\mathcal{P}(\text{Col}(\omega, \sigma))^{L[x, T]}$ is countable in V .
- (10.5) We conclude that $\mathcal{P}(\text{Col}(\omega, \sigma))^{W_\sigma}$ is countable as well and that there exists $g \rightsquigarrow W_\sigma^{\text{Col}(\omega, \sigma)}$.
- (10.6) Since $\mathbb{R}^{W_\sigma} = \sigma$, [2.0.9] implies $\omega_1^{W_\sigma[g]} \leq \Theta^{W_\sigma}$.
- (10.7) Let $y \in W_\sigma[g]$ be a real coding g .
- (10.8) Let $g_0 \in W_\sigma[g]$ be a real coding g . We have

$$W_\sigma[g] = L(\sigma; T, g) = L(g_0; T) = L[g_0, T] \models \text{ZFC}.$$

- (10.9) Let $y \in \mathbb{R}^{W_\sigma[g]}$ be such that $\sigma = \sigma_y$.
- (10.10) There is $y' \in \mathbb{R}$ such that $\alpha_{y'} = f(\sigma)$; we have $(y, y') \in A = p[T]$.
- (10.11) Thus, T_y has a branch in V .
- (10.12) By [1.1.8], T_y has a branch in $W_\sigma[g]$.
- (10.13) Let (z, w) be one such branch. We have $z \in \mathbb{R}^{W_\sigma[g]}$ and $(y, z) \in p[T] = A$.
- (10.14) This yields: $f(\sigma) = f(\sigma_y) = \alpha_z < \omega_1^{W_\sigma[g]} \leq \Theta^{W_\sigma}$.

□

- (11) By (10), we have $[f] < [\sigma \mapsto \Theta^{W_\sigma}] = \Theta^W \leq \Theta$.

□

3 $\text{Col}(\omega, \mathbb{R})$ -generic Ultrapowers

3.1 Description of the Ultrafilter

[3.1.1] **Declaration.** We assume in this section that $V = L(\mathcal{P}(\mathbb{R})) \models \text{AD}_{\mathbb{R}} + (\Theta \text{ regular})$. We denote by μ the club ultrafilter.

[3.1.2] **Remark.** Recall that $\mathcal{F}_{\text{Col}(\omega, X)}$ is naturally identified with X^ω (with the Tychonoff topology built from the discrete topology on X).

[3.1.3] **Definition.** Let $A \subseteq \mathbb{R}^\omega$.

- a. Set A is *weakly co-meager below* $p \in \text{Col}(\omega, \mathbb{R})$ if there exists a club $C \subseteq [\mathbb{R}]^\omega$ such that for every $\sigma \in C$ we have $\text{im}(p) \subseteq \sigma$ and set $A \cap \sigma^\omega \cap \mathcal{N}_p$ is co-meager in $\mathcal{N}_p \subseteq \mathcal{F}_{\text{Col}(\omega, \sigma)} = \sigma^\omega$.
- b. A is *weakly co-meager* if it is weakly co-meager below $\emptyset$.

[3.1.4] **Remark.** If $q \leq p$ and A is weakly co-meager below p , then A is weakly co-meager below q .

[3.1.5] **Definition.** Let $g \rightsquigarrow V^{\text{Col}(\omega, \mathbb{R})}$. Then we define $U_g \subseteq \mathcal{P}(\mathbb{R}^\omega)^V$ as follows: $A \in U_g$ if and only if there exists some $p \in g$ such that $V \models (A \text{ is weakly co-meager below } p)$.

[3.1.6] **Lemma.** For every $A \subseteq \mathbb{R}$, either A is co-meager or there exists $p \in \omega^{<\omega}$ such that $\mathcal{N}_p - A$ is co-meager in $\mathcal{N}_p$.

Proof.

- (1) By AD, there exists an open $U \subseteq \mathbb{R}$ such that $U \triangle A$ is meager.
- (2) If $U = \mathbb{R}$, then A is co-meager.
- (3) Otherwise, $U \neq \mathbb{R}$ and there exists $p \in \omega^{<\omega}$ such that $\mathcal{N}_p \cap U = \emptyset$.
- (4) Then $\mathcal{N}_p - (\mathcal{N}_p - A) \subseteq \mathcal{N}_p \cap A \subseteq A - U \subseteq A \triangle U$ and this last set is meager. Hence, $\mathcal{N}_p - A$ is co-meager in $\mathcal{N}_p$.

□

[3.1.7] **Lemma.** Let $A \subseteq \mathbb{R}$ and let $p \in \text{Col}(\omega, \mathbb{R})$. Then either A is weakly co-meager below p or there exists $q \leq p$ such that $\mathbb{R}^\omega - A$ is weakly co-meager below q .

Proof.

- (1) Suppose that A is not weakly co-meager below p .
- (2) Then there exists stationary $S \subseteq [\mathbb{R}]^\omega$ such that for every $\sigma \in S$ it holds that:
 - a. the code for every $s \in \sigma^{<\omega}$ belongs to σ ;
 - b. $\text{im}(p) \subseteq \sigma$
 - c. $A \cap \sigma^\omega \cap \mathcal{N}_p$ is not co-meager in $\mathcal{N}_p$.
- (3) By [3.1.6], for every $\sigma \in S$, there exists $q \in \text{Col}(\omega, \sigma)$ such that $q \leq p$ and $\mathcal{N}_q - A$ is co-meager.
- (4) Let $f : S \rightarrow [\mathbb{R}]^\omega$ be defined by

$$f(\sigma) := \{x \in \sigma : x \text{ codes some } q \in \sigma^{<\omega}, q \leq p, \mathcal{N}_q - A \text{ is co-meager}\}.$$

- (5) It holds that $\emptyset \subset f(\sigma) \subseteq \sigma$ for every $\sigma \in S$.
- (6) By normality there exist stationary $S' \subseteq S$ and $x \in \mathbb{R}$ such that $x \in \cap f[S']$.
- (7) Let $q \in \text{Col}(\omega, \mathbb{R})$ be the condition coded by x . We have $q \leq p$.
- (8) For every $\sigma \in S'$, we have $\text{im}(p) \subseteq \sigma$ and $\mathcal{N}_q - A$ is co-meager.
- (9) By [1.5.13], we have that S' is club, which yields that $\mathbb{R}^\omega - A$ is co-meager below q .

□

[3.1.8] **Proposition.** Let $g \rightsquigarrow V^{\text{Col}(\omega, \mathbb{R})}$. Then U_g is a V -ultrafilter on $(\mathbb{R}^\omega)^V$.

Proof.

- (1) It is readily verifiable that U_g is a filter. Let us verify that it is maximal.
- (2) Suppose $A \notin U_g$. Then for every $p \in g$, set A is not weakly co-meager below p .
- (3) Let $D := \{p \in \text{Col}(\omega, \mathbb{R}) : A \text{ is weakly co-meager below } p\} \cup \{p \in \text{Col}(\omega, \mathbb{R}) : A^\mathbb{G} \text{ is weakly co-meager below } p\}$. By [3.1.7], set D is dense in $\text{Col}(\omega, \mathbb{R})$.
- (4) Let $p \in g \cap D$.
- (5) By (2), set A is not weakly co-meager below p . Thus, $A^\mathbb{G}$ is weakly co-meager below $p \in g$.
- (6) We conclude $A^\mathbb{G} \in U_g$.

□

[3.1.9] **Notation.** Let us denote $\pi : \mathbb{R}^\omega \rightarrow [\mathbb{R}]^\omega : \sigma \mapsto \text{im}(\sigma)$ and let μ be the club ultrafilter on $[\mathbb{R}]^\omega$.

[3.1.10] **Lemma.** For every $\sigma \in [\mathbb{R}]^\omega$, set $\{u \in \sigma^\omega : \text{im}(u) = \sigma\}$ is co-meager in σ^ω .

Proof.

- (1) Let $x \in \sigma$ be arbitrary and let us verify that $U_x := \{u \in \sigma^\omega : x \in \text{im}(u)\}$ is dense and open. Since σ is countable, this suffices for the conclusion.
- (2) To verify density, let $p \in \sigma^{<\omega}$ be arbitrary. Then $p \frown x \in U_x \cap \mathcal{N}_p$.
- (3) To verify openness, let $u \in U_x$ be arbitrary.
- (4) Then there exists $n < \omega$ such that $x \in \text{im}(u \upharpoonright n)$.
- (5) Hence, $x \in \mathcal{N}_{u \upharpoonright n} \subseteq U_x$.

□

[3.1.11] **Lemma.** Let $g \rightsquigarrow V^{\text{Col}(\omega, \mathbb{R})}$. Then $\pi_* U_g = \mu$.

Proof.

- (1) Let $A \in \mathcal{P}([\mathbb{R}]^\omega)^V$ be arbitrary. We want to verify that $A \in \mu$ if and only if $\pi^{-1}[A] \in U_g$.
- (2) Suppose first that $A \in \mu$. It suffices to verify that $\pi^{-1}[A]$ is weakly co-meager.
- (3) $C := A \subseteq [\mathbb{R}]^\omega$ is club and for every $\sigma \in C$, we have by [3.1.10] that $\{u \in \sigma^\omega : \text{im}(u) = \sigma\}$ is co-meager in σ^ω .
- (4) Hence, for every $\sigma \in C$, set $\pi^{-1}[A] \cap \sigma^\omega \supseteq \{u \in \sigma^\omega : \text{im}(u) = \sigma\}$ is co-meager in σ^ω . This establishes that $\pi^{-1}[A]$ is weakly co-meager.
- (5) Suppose now that $\pi^{-1}[A] \in U_g$. We want to verify that A is club.
- (6) There exists $p \in g$ and club $C \subseteq [\mathbb{R}]^\omega$ such that for every $\sigma \in C$, we have $\text{im}(p) \subseteq \sigma$ and $\pi^{-1}[A] \cap \sigma^\omega \cap \mathcal{N}_p$ is co-meager in $\mathcal{N}_p$.
- (7) Let $\sigma \in C$ be arbitrary. By [3.1.10], there exists $u \in \sigma^\omega$ such that $\text{im}(u) = \sigma$ and $u \in \pi^{-1}[A] \cap \sigma^\omega$.
- (8) Then $\sigma = \text{im}(u) = \pi(u) \in A$.
- (9) Thus, $C \subseteq A$ and we conclude that A is club.

□

3.2 The Ultrapower

[3.2.1] **Remark.** We can form ultrapower $\text{Ult}(V, U_g)$ whenever $g \rightsquigarrow V^{\text{Col}(\omega, \mathbb{R})}$. Since $V \not\models \text{AC}$, we need to verify Łoś's theorem ad hoc.

[3.2.2] **Proposition.** Suppose that $g \rightsquigarrow V^{\text{Col}(\omega, \mathbb{R})}$. Let $\phi(x_0, \dots, x_{n-1})$ be a formula and let $f_0, \dots, f_{n-1} \in (\mathbb{R}^\omega \rightarrow V)^V$. Then

$$\text{Ult}(V, U_g) \models \phi[[f_0], \dots, [f_{n-1}]] \iff \{\sigma \in \mathbb{R}^\omega : V \models \phi[f_0(\sigma), \dots, f_{n-1}(\sigma)]\} \in U_g.$$

Proof.

- (1) Let us fix surjection $F : \mathcal{P}(\mathbb{R}) \times \text{Ord} \rightarrow V$.
- (2) We prove the proposition by induction on complexity of ϕ .
- (3) Only implication ($\Leftarrow$) of the inductive case $\phi = \exists y \psi$ is non-trivial, so that is what we consider here.
- (4) Suppose that $A \in U_g$ and for every $\sigma \in A$, we have

$$V \models \exists y \psi(f(\sigma), y).$$

- (5) We reason now in V .
- (6) For $\sigma \in A$, let $(\eta_\sigma, \alpha_\sigma)$ be the lexicographically minimal pair $(\eta, \alpha) \in \Theta \times \text{Ord}$ such that there exists $B \in \mathcal{P}_\eta(\mathbb{R})$ satisfying

$$V \models \psi[f(\sigma), F(B, \alpha)].$$

- (7) Since $A \subseteq \mathbb{R}^\omega$, we have

$$\eta := \sup\{\eta_\sigma : \sigma \in A\} < \Theta.$$

- (8) Let $R \subseteq \mathbb{R}^\omega \times \mathcal{P}_\eta(\mathbb{R})$ be defined by: $(\sigma, B) \in R$ if and only if $\sigma \in A$ and $V \models \psi[f(\sigma), F(B, \alpha_\sigma)]$.
- (9) Since elements of $\mathbb{R}^\omega$ and elements of $\mathcal{P}_\eta(\mathbb{R})$ can be coded by reals, we may apply [1.5.3] and obtain $h : \mathbb{R}^\omega \rightarrow \mathcal{P}_\eta(\mathbb{R})$ such that $(\sigma, h(\sigma)) \in R$ for every $\sigma \in A$.
- (10) Let $u : A \rightarrow V$ be defined by $u(\sigma) := F(h(\sigma), \alpha_\sigma)$. For every $\sigma \in A$, we have $V \models \psi[f(\sigma), u(\sigma)]$.
- (11) We reason again in $V[g]$. Since $A \in U_g$, the IH implies $\text{Ult}(V, U_g) \models \psi[[f], [u]]$.

□

[3.2.3] **Corollary.** For $g \rightsquigarrow V^{\text{Col}(\omega, \mathbb{R})}$, mapping $j_g : V \rightarrow \text{Ult}(V, U_g) : x \mapsto [\sigma \mapsto x]_{U_g}$ is correctly defined and elementary. □

[3.2.4] **Lemma.** Let γ be an ordinal and let $(A_\alpha : \alpha < \gamma)$ be a family of meager sets in $\mathbb{R}$. Then $\bigcup_{\alpha < \gamma} A_\alpha$ is meager.

Proof. This is a well-known AD theorem. □

[3.2.5] **Proposition.** Let $g \rightsquigarrow V^{\text{Col}(\omega, \mathbb{R})}$ and let

$$k : \text{Ult}(V, \mu) \mapsto \text{Ult}(V, U_g) : [f]_\mu \mapsto [f \circ \pi]_{U_g}.$$

Then k is correctly defined, it is an isomorphism onto its image, and it satisfies $k \circ j_\mu = j_g$ and $k \upharpoonright \text{Ord} = \text{id}$. In particular, $\text{Ult}(V, U_g)$ is well-founded.

Proof.

- (1) The fact that k is correctly defined and that it is an isomorphism onto its image follows from the previous lemma. The fact that $k \circ j_\mu = j_g$ is obvious.
- (2) We have now that

$$k \upharpoonright \text{Ord}^{\text{Ult}(V, \mu)} : \text{Ord}^{\text{Ult}(V, \mu)} \rightarrow \text{Ord}^{\text{Ult}(V, U_g)}$$

is an injection which agrees with the order. We will be done if we show that it is surjective.

- (3) Let $f \in (\mathbb{R}^\omega \rightarrow \mathbf{Ord})^V$ and let us show that there is $h \in ([\mathbb{R}]^\omega \rightarrow \mathbf{Ord})^V$ such that $\{\sigma \in \mathbb{R}^\omega : f(\sigma) = h(\text{im}(\sigma))\} \in U_g$.
- (4) We want to find $h \in ([\mathbb{R}]^\omega \rightarrow \mathbf{Ord})^V$ such that there exists $p \in g$ such that $\{\sigma \in \mathbb{R}^\omega : f(\sigma) = h(\text{im}(\sigma))\}$ is weakly co-meager below p .
- (5) Inside V , let D be the set of all $p \in \text{Col}(\omega, \mathbb{R})$ for which there exists $h : [\mathbb{R}]^\omega \rightarrow \mathbf{Ord}$ such that $\{\sigma \in \mathbb{R}^\omega : f(\sigma) = h(\text{im}(\sigma))\}$ is weakly co-meager below p . It suffices to show that D is dense.
- (6) Let $p \in D$ and let us find $q \in D$ such that $q \leq p$.
- (7) Let C be the club of infinite elements of $[\mathbb{R}]^\omega$ which are closed under coding $\mathbb{R}^{<\omega} \rightarrow \mathbb{R}$ and let $\gamma := \sup \text{im}(f) + 1$.
- (8) Let $\sigma \in C$. We have

$$\mathbb{R} \approx \sigma^\omega \cap \mathcal{N}_p = \bigcup_{\alpha < \gamma} \{\tau \in \sigma^\omega : p \triangleleft \tau \wedge f(\tau) = \alpha\}.$$

By the previous lemma, there exists $\alpha < \gamma$ such that $\{\tau \in \sigma^\omega : p \triangleleft \tau \wedge f(\tau) = \alpha\}$ is not meager in $\mathcal{N}_p$. We define $h(\sigma)$ to be the minimal such α .

- (9) We want to find $q \leq p$ such that $\{\sigma \in \mathbb{R}^\omega : f(\sigma) = h(\text{im}(\sigma))\}$ is weakly co-meager below q .
- (10) Given $\sigma \in C$, set $\{\tau \in \sigma^\omega : f(\tau) = h(\sigma)\} \cap \mathcal{N}_p$ is not meager in $\mathcal{N}_p$, so there exists $q \leq p$ such that $\{\tau \in \sigma^\omega : f(\tau) = h(\sigma)\} \cap \mathcal{N}_q$ is co-meager in $\mathcal{N}_q$.
- (11) Since the set of all $\tau \in \sigma^\omega$ such that $\text{im}(\tau) = \sigma$ is co-meager in σ^ω , we conclude that for some $q \leq p$ set $\{\tau \in \sigma^\omega : f(\tau) = h(\text{im}(\tau))\} \cap \mathcal{N}_q$ is co-meager in $\mathcal{N}_q$. Note that necessarily such q belongs to $\sigma^{<\omega}$, so it is coded by some $x \in \sigma$.
- (12) Consider the function $C \rightarrow [\mathbb{R}]^\omega$ defined by mapping $\sigma \in C$ to the set of all $x \in \sigma$ which code some $q \in \text{Col}(\omega, \mathbb{R})$ satisfying that $q \leq p$ and set $\{\tau \in \sigma^\omega : f(\tau) = h(\text{im}(\tau))\} \cap \mathcal{N}_q$ is co-meager in $\mathcal{N}_q$. We may now apply normality of μ , which yields a club $C' \subseteq C$ and a real x that codes some $q \in \text{Col}(\omega, \mathbb{R})$ such that $q \leq p$ and for every $\sigma \in C'$, we have that set $\{\tau \in \sigma^\omega : f(\tau) = h(\text{im}(\tau))\} \cap \mathcal{N}_q$ is co-meager in $\mathcal{N}_q$.
- (13) In other words, $\{\sigma \in \mathbb{R}^\omega : f(\sigma) = h(\text{im}(\sigma))\}$ is weakly co-meager below q , implying $q \in D$.

□

[3.2.6] **Corollary.** Let A be a set of ordinals in V and let $g \rightsquigarrow V^{\text{Col}(\omega, \mathbb{R})}$. Then $j_g(A) = j_\mu(A)$.

Proof. For an ordinal α , we reason as follows:

$$\alpha \in j_g(A) \iff k(\alpha) \in k(j_\mu(A)) \iff \alpha \in j_\mu(A).$$

□

[3.2.7] **Lemma.** Let $g \rightsquigarrow V^{\text{Col}(\omega, \mathbb{R})}$ and let $f \in ([\mathbb{R}]^\omega \rightarrow \mathbb{R})^V$. Then $[f]_\mu = [f \circ \pi]_{U_g}$.

Proof.

- (1) We want to show that for every $m, n < \omega$, it holds that

$$[f]_\mu(m) = n \iff [f \circ \pi]_{U_g}(m) = n.$$

- (2) If $[f \circ \pi]_{U_g}(m) = n$, then

$$\{\sigma \in (\mathbb{R}^\omega)^V : f(\text{im}(\sigma))(m) = n\} \in U_g.$$

- (3) This means

$$\pi^{-1}[\{\sigma \in ([\mathbb{R}]^\omega)^V : f(\sigma)(m) = n\}] \in U_g$$

and also

$$\{\sigma \in ([\mathbb{R}]^\omega)^V : f(\sigma)(m) = n\} \in \mu$$

by [3.1.11].

- (4) Hence, $[f]_\mu(m) = n$.
(5) On the other hand, if $[f \circ \pi]_{U_g}(m) \neq n$, then

$$\{\sigma \in (\mathbb{R}^\omega)^V : f(\text{im}(\sigma))(m) \neq n\} \in U_g.$$

- (6) Analogously as above,

$$\{\sigma \in ([\mathbb{R}^\omega])^V : f(\sigma)(m) \neq n\} \in \mu,$$

i.e. $[f]_\mu(m) \neq n$.

□

[3.2.8] **Proposition.** Let $g \rightsquigarrow V^{\text{Col}(\omega, \mathbb{R})}$. Then:

- a. $[\text{id}_{\mathbb{R}^\omega}]_{U_g} = \bigcup g$
- b. $\mathbb{R}^V = [\pi]_{U_g} \in \text{Ult}(V, U_g)$
- c. for every $f \in ([\mathbb{R}]^\omega \rightarrow V)^V$, it holds that $[f]_\mu = j_g(f)(\mathbb{R}^V)$.

Proof.

- (1) We first show [a](#). Let $n < \omega$ and let us show that $[\text{id}_{\mathbb{R}^\omega}]_{U_g}(n) = \bigcup g(n)$.
- (2) Let $p \in g$ be such that $n \in \text{dom}(p)$ and let $x := p(n)$. We want to show that $[\text{id}_{\mathbb{R}^\omega}]_{U_g}(n) = x$.
- (3) It suffices to verify that $\{\sigma \in (\omega \rightarrow \mathbb{R}) \cap V : \sigma(n) = x\}$ is weakly co-meager below p .
- (4) In other words, we want to find a club $C \subseteq [\mathbb{R}]^\omega$ such that for every $\sigma \in C$, it holds that $p \in \sigma^{<\omega}$ and that set $A_\sigma := \{\tau \in (\omega \rightarrow \sigma) \cap V : \tau(n) = x\}$ is co-meager in $\mathcal{N}_p$.
- (5) We may choose $C := \{\sigma \in [\mathbb{R}]^\omega : \text{im}(p) \subseteq \sigma\}$, since for every $\sigma \in C$, we have $A_\sigma \cap \mathcal{N}_p = \mathcal{N}_p$.
- (6) Part [b](#) is now deduced as follows:

$$\mathbb{R}^V = \text{im}(\bigcup g) = \text{im}[\text{id}]_{U_g} = [(\text{im}(\sigma))_\sigma] = [\pi].$$

- (7) For [c](#), reason as follows:

$$[f]_\mu = [f \circ \pi]_{U_g} = j_g(f)([\pi]_{U_g}) = j_g(f)(\mathbb{R}^V),$$

where the first equality follows from [\[3.2.7\]](#) and the last follows from [b](#).

□

[3.2.9] **Proposition.** Let $g \rightsquigarrow V^{\text{Col}(\omega, \mathbb{R})}$. Then $\mathbb{R}^{V[g]} \subseteq \text{Ult}(V, U_g)$.

Proof.

- (1) Let $\dot{x}$ be a $\text{Col}(\omega, \mathbb{R})$ -name for a real. We want to show $\dot{x}^g \in \text{Ult}(V, U_g)$.
- (2) For $n, i < \omega$, let $D_{n,i}$ be the set of all conditions p such that $p \Vdash \dot{x}(n) = i$.
- (3) For every $\sigma \in \mathbb{R}^\omega$ and every $n < \omega$, there exists unique $i < \omega$ such that

$$\{p \in D_{n,i} : p \triangleleft \sigma\} \neq \emptyset.$$

Let $F(\sigma)(n)$ be that i . We want to show that $[F]_{U_g} = \dot{x}^g$.

- (4) Let $n < \omega$ be arbitrary and let us verify that $[F]_{U_g}(n) = \dot{x}^g(n)$.
- (5) Let $p \in g$ and $i < \omega$ be such that $p \Vdash \dot{x}(n) = i$. We are done if we show that $\{\sigma \in (\mathbb{R}^\omega)^V : F(\sigma)(n) = i\}$ is weakly co-meager below p .
- (6) We work now inside V . Let $\sigma \in [\mathbb{R}]^\omega$ be arbitrary and let us show that

$$\{\tau \in \sigma^\omega : p \triangleleft \tau \wedge F(\tau)(n) = i\}$$

is co-meager in $\mathcal{N}_p \subseteq \sigma^\omega$.

- (7) Since $p \Vdash \dot{x}(n) = i$, we have $p \in D_{n,i}$. This means that if $\tau \in \mathbb{R}^\omega$ and $p \triangleleft \tau$, we have $F(\tau)(n) = i$. We conclude that

$$\{\tau \in \sigma^\omega : p \triangleleft \tau \wedge F(\tau)(n) = i\} = \mathcal{N}_p$$

and hence it is co-meager. □

[3.2.10] **Definition.**

- a. Tree T on $\omega \times \gamma$ is $\mathbb{R}$ -*absolutely complemented* if there exists a tree U on $\omega \times \delta$ such that

$$V^{\text{Col}(\omega, \mathbb{R})} \models p[T] \sqcup p[U] = \mathbb{R}.$$

- b. Set $A \subseteq \mathbb{R}$ is $\mathbb{R}$ -*universally Baire* if there exists an $\mathbb{R}$ -absolutely complemented tree T such that $p[T] = \mathbb{R}$.
c. Let $A \subseteq \mathbb{R}$ be $\mathbb{R}$ -universally Baire and let $g \rightsquigarrow V^{\text{Col}(\omega, \mathbb{R})}$. We define $A^g := p[T]^{V[g]}$.

[3.2.11] **Remark.** The usual argument shows that A^g is correctly defined.

[3.2.12] **Proposition.** Let T be a tree on $\omega \times \gamma$.

- a. Then $T^* := j_\mu(T)$ is a $\mathbb{R}$ -absolutely complemented tree on $\omega \times j_\mu(\gamma)$ and $p[T^*] = p[T]$.
b. If $A := p[T]$ and $g \rightsquigarrow V^{\text{Col}(\omega, \mathbb{R})}$, then A is $\mathbb{R}$ -universally Baire and $A^g = j_g(A)$.

Proof.

- (1) The fact that $p[T^*] = p[T]$ is obvious.
- (2) Since every set is Suslin, there exists tree U such that $p[U] = p[T]^{\mathbb{C}}$. Let $U^* := j_\mu(U)$.
- (3) $V \models p[T] \sqcup p[U] = \mathbb{R}$
- (4) $\text{Ult}(V, U_g) \models p[j_g(T)] \sqcup p[j_g(U)] = \mathbb{R}$.
- (5) Since $\mathbb{R}^{\text{Ult}(V, U_g)} = \mathbb{R}^{V[g]}$, we conclude

$$V[g] \models p[j_g(T)] \sqcup p[j_g(U)] = \mathbb{R}.$$

- (6) By [3.2.6], we have $j_g(T) = T^*$ and $j_g(U) = U^*$. We are done with the first part.
- (7) For the second part, we have

$$A^g = p[T^*]^{V[g]} = p[j_g(T)]^{V[g]} = p[j_g(T)]^{\text{Ult}(V, U_g)} = j_g(p[T]^V) = j_g(A).$$

□

[3.2.13] **Corollary.** Every set of reals is $\mathbb{R}$ -universally Baire.

Proof. It follows from the previous proposition and the fact that every set of reals is Suslin. □

4 Forcing Ideals from Determinacy

4.1 Fine-structural Lemmas

[4.1.1] **Summary.** We work under the assumption that $V = L(X)$ for transitive set X . The aim is to establish existence of a certain transitive model of set theory.

[4.1.2] **Lemma.** Regular cardinals are cofinal in Ord .

Proof. Let $g \rightsquigarrow V^{\text{Col}(\omega, X)}$; then $V[g] \models \text{ZFC}$ and consequently regular cardinals are cofinal in Ord . Those cardinals remain regular in $L(X)$, which suffices for the conclusion. $\square$

[4.1.3] **Lemma.** Let $\theta \geq \Theta_{X^{<\omega}}$ be a regular cardinal. Then $J_\theta(X) \models \text{ZF}^-$.

Proof.

(1) We need only to verify the collection schema. Let $R \subseteq J_\theta(X)^2$ be arbitrary satisfying

$$(\forall x \in J_\theta(X))(\exists y \in J_\theta(X))((x, y) \in R)$$

and let $u \in J_\theta(X)$. We want to find $v \in J_\theta(X)$ such that

$$(\forall x \in u)(\exists y \in v)((x, y) \in R)$$

(2) There exists a surjection $F : X^{<\omega} \times \theta \rightarrow J_\theta(X)$ such that for every limit $\eta < \theta$, we have $F[X^{<\omega} \times \eta] = J_\eta(X)$.

(3) There exists a limit $\xi < \theta$ such that $u \subseteq J_\xi(X) = F[X^{<\omega} \times \xi]$.

(4) Given $s \in X^{<\omega}$ and $\alpha < \xi$, let $\beta_{s,\alpha}$ be the minimal $\beta < \theta$ with the property

$$(\exists t \in X^{<\omega})((F(s, \alpha), F(t, \beta)) \in R).$$

(5) Note that for every $\alpha < \xi$, we have $|\{\beta_{s,\alpha} : s \in X^{<\omega}\}| < \Theta_{X^{<\omega}} \leq \theta$.

(6) Since θ is regular, we conclude

$$\gamma_\alpha := \sup\{\beta_{s,\alpha} : s \in X^{<\omega}\} < \theta.$$

(7) Since $\xi < \theta$ and θ is regular, we conclude

$$\delta := \sup_{\alpha < \xi} (\gamma_\alpha + 1) < \theta.$$

(8) For every $v := F[X^{<\omega} \times \delta] = J_\delta(X)$ is as required. $\square$

[4.1.4] **Lemma.** Suppose that DC holds. Let $\theta \geq \Theta_{X^{<\omega}}$ be a regular cardinal and let $f : X^{<\omega} \rightarrow J_\theta(X)$. Then there exists $N \prec J_\theta(X)$ such that $\text{im} f \subseteq N$ and there is a surjection $X^{<\omega} \rightarrow N$.

Proof.

(1) Let us fix a surjection $U : X^{<\omega} \times \theta \rightarrow J_\theta(X)$.

(2) Given a formula $\phi = \phi(\bar{x}, y)$ and an $\bar{x}$ -tuple $\bar{a} \in J_\theta(X)$ such that $J_\theta(X) \models \exists y \phi(\bar{a}, y)$, let $\alpha_{\phi, \bar{a}}$ be the minimal ordinal $\alpha < \theta$ such that

$$\{s \in X^{<\omega} : J_\theta(X) \models \phi[\bar{a}, U(s, \alpha)]\} \neq \emptyset;$$

if $J_\theta(X) \models \neg \exists y \phi(\bar{a}, y)$, we set $\alpha_{\phi, \bar{a}} = 0$.

(3) By recursion on $n < \omega$, we define sets $N_n \subseteq J_\theta(X)$.

a. $N_0 := \text{im} f$

- b. $N_{n+1} := N_n \cup \{U(s, \alpha_{\phi, t}) : \phi(\bar{x}, y) \text{ is a formula, } t \in N_n^{|\bar{x}|}, s \in X^{<\omega}\}$.
- (4) By Tarski-Vaught test, we conclude

$$N := \bigcup_{n < \omega} N_n \prec J_\theta(X).$$

- (5) It is easily established by induction that for every $n < \omega$, there exists a surjection $X^{<\omega} \rightarrow N_n$.
- (6) Since DC holds, we can choose a sequence $F_n : X^{<\omega} \rightarrow N_n$ of surjections for $n < \omega$. This means that there exists a surjection $X^{<\omega} \rightarrow N$.

□

[4.1.5] **Lemma.** Suppose that DC holds. Let $\theta \geq \Theta_{X^{<\omega}}$ be regular. Then $J_\theta(X) \models \text{DC}$.

Proof.

- (1) By [4.1.3], we have $J_\theta(X) \models \text{ZF}^-$. In particular, we have a surjection $F : X^{<\omega} \times \theta \rightarrow J_\theta(X)$ and the proof of [1.1.6] in the case $\lambda = \omega$ goes through.
- (2) In other words, it suffices to establish that every tree without leaves on $X^{<\omega}$ that is inside $J_\theta(X)$ has a branch in $J_\theta(X)$.
- (3) Since $(X^{<\omega})^{<\omega}$ can be coded by $X^{<\omega}$, it suffices in turn to consider trees on X . Since DC holds in V , we will be done if we show $X^\omega \subseteq J_\theta(X)$.
- (4) Let $x \in X^\omega$ be arbitrary and let $\chi > \theta$ be regular such that $x \in J_\chi(X)$. By [4.1.4], there is some $N \prec J_\chi(X)$ such that $X \cup \{x\} \subseteq N$ and there is surjection $F : X^{<\omega} \rightarrow N$.
- (5) Let P be the collapse of N . Since $X \in P$ and $N \prec J_\chi(X)$, we have $P = J_\alpha(X)$ for some α .
- (6) Since there is a surjection $X^{<\omega} \rightarrow P$, we conclude $\alpha < \Theta_{X^{<\omega}} \leq \theta$.
- (7) Hence, $x \in P \subseteq J_\theta(X)$.

□

[4.1.6] **Lemma.** Suppose $V = L(\mathcal{P}(\mathbb{R}))$, that $\text{AD}_{\mathbb{R}}$ holds, and that Θ is regular. Let $A \subseteq \mathbb{R}$. Then there exists transitive model N such that:

- a. $N \models \text{ZF}^- + \text{DC}$
- b. $\mathbb{R} \subseteq N$ and $A \in N$;
- c. $N \models (A \text{ is } \text{Col}(\omega, \mathbb{R})\text{-universally Baire})$
- d. there exists a surjection $F : \mathbb{R} \rightarrow N$.

Proof.

- (1) By [4.1.2], there exists a regular $\theta \geq \Theta_{\mathcal{P}(\mathbb{R})}$ such that $J_\theta(\mathcal{P}(\mathbb{R}))$ contains a pair of trees T, S witnessing that A is $\text{Col}(\omega, \mathbb{R})$ -universally Baire.
- (2) By [4.1.3], we have $J_\theta(\mathcal{P}(\mathbb{R})) \models \text{ZF}^-$; by [4.1.5], we have $J_\theta(\mathcal{P}(\mathbb{R})) \models \text{DC}$; by (1), we have $J_\theta(\mathcal{P}(\mathbb{R})) \models (A \text{ is } \text{Col}(\omega, \mathbb{R})\text{-universally Baire})$.
- (3) Let us fix a surjection $U : \mathcal{P}(\mathbb{R}) \times \theta \rightarrow J_\theta(\mathcal{P}(\mathbb{R}))$.
- (4) Given a formula $\phi = \phi(\bar{x}, y)$ and an $\bar{x}$ -tuple $\bar{a} \in J_\theta(\mathcal{P}(\mathbb{R}))$ such that $J_\theta(\mathcal{P}(\mathbb{R})) \models \exists y \phi(\bar{a}, y)$, let $(\alpha_{\phi, \bar{a}}, \eta_{\phi, \bar{a}})$ be the lexicographically minimal pair of ordinals $(\alpha, \eta) \in (\theta, \Theta)$ such that

$$\{B \in \mathcal{P}_\eta(\mathbb{R}) : J_\theta(\mathcal{P}(\mathbb{R})) \models \phi[\bar{a}, U(B, \alpha)]\} \neq \emptyset;$$

if $J_\theta(\mathcal{P}(\mathbb{R})) \models \neg \exists y \phi(\bar{a}, y)$, we set $\alpha_{\phi, \bar{a}} = \eta_{\phi, \bar{a}} = 0$.

- (5) By recursion on $n < \omega$, we define sets $N_n \subseteq J_\theta(\mathcal{P}(\mathbb{R}))$.
 - a. $N_0 := \mathbb{R} \cup \{A\}$
 - b. $\eta_n := \sup\{\eta_{\phi, s} : \phi(\bar{x}, y) \text{ is a formula, } s \in N_n^{|\bar{x}|}\}$

c. $N_{n+1} := N_n \cup \{U(B, \alpha_{\phi, s}) : \phi(\bar{x}, y) \text{ is a formula, } s \in N_n^{|\bar{x}|}, B \in \mathcal{P}_{\eta_n}(\mathbb{R})\}$.

(6) By Tarski-Vaught test, we conclude

$$N_\omega := \bigcup_{n < \omega} N_n \prec J_\theta(\mathcal{P}(\mathbb{R})).$$

(7) **Claim.** For every $n < \omega$, there exists a surjection $F : \mathbb{R} \rightarrow N_n$.

Proof.

(7.1) We reason by induction. Case $n = 0$ is obvious.

(7.2) Consider inductively the case of index $n + 1$.

(7.3) Since there exists a surjection $\mathbb{R} \rightarrow N_n$, we can construct a surjection $f : \mathbb{R} \rightarrow \{(\phi, s) : \phi(\bar{x}, y) \text{ is a formula, } s \in N_n^{|\bar{x}|}\}$.

(7.4) Thus, $\eta_n = \sup\{\eta_{f(x)} : x \in \mathbb{R}\}$. Since $\eta_{f(x)} < \Theta$ for every $x \in \mathbb{R}$, $|\{\eta_{f(x)} : x \in \mathbb{R}\}| < \Theta$, and Θ is regular, we conclude $\eta_n < \Theta$.

(7.5) Let $B_n \subseteq \mathbb{R}$ be of Wadge rank η_n . We have now $N_{n+1} = N_n \cup \{U(u^{-1}[B_n], \alpha_{\phi, s}) : \phi(\bar{x}, y) \text{ is a formula, } s \in N_n^{|\bar{x}|}, u : \mathbb{R} \rightarrow \mathbb{R} \text{ continuous}\}$.

(7.6) By the IH and the fact that continuous functions are coded by reals, we conclude that there is a surjection $F : \mathbb{R} \rightarrow N_{n+1}$. □

(8) Since DC holds, we can choose a sequence $F_n : \mathbb{R} \rightarrow N_n$ of surjections for $n < \omega$. This means that there exists a surjection $\mathbb{R} \rightarrow N_\omega$.

(9) Let N be the transitive collapse of N_ω ; this N is as required. □

4.2 Forcing Properties on $\text{NS}_{[\mathbb{R}]^\omega}$

[4.2.1] **Declaration.** We work for the rest of the section under the assumption that $V = L(\mathcal{P}(\mathbb{R}))$, that $\text{AD}_{\mathbb{R}}$ holds, and that Θ is regular. We fix notation μ for the club ultrafilter on $[\mathbb{R}]^\omega$.

[4.2.2] **Summary.** We want to show that inside $(V^{\text{Col}(\omega_1, \mathbb{R})})^{\text{Col}(\omega_2, \mathcal{P}(\mathbb{R}))}$, ideal $\text{NS}_{[\mathbb{R}]^\omega}$ is strong and pseudo-homogeneous. Our strategy for the proof is the following; we speak from the point of view of $V^{\text{Col}(\omega_1, \mathbb{R})}$. We will show that $\text{NS}_{[\mathbb{R}]^\omega}$ has the ordinal covering property w.r.t. V . From this we will conclude that $V[\dot{G}]^{\text{Col}(\omega_2, \mathbb{R})} \models (\text{NS}_{[\mathbb{R}]^\omega} \text{ has the ordinal covering property w.r.t. } V)$. To reach the final conclusion, we show that inside $V[\dot{G}]^{\text{Col}(\omega_2, \mathbb{R})}$ the ordinal covering property w.r.t. V implies strongness and pseudo-homogeneity.

[4.2.3] **Proposition.**

a. $\text{Col}(\omega_1, \mathbb{R})$ is ω -closed and homogeneous².

Let $g \rightsquigarrow V^{\text{Col}(\omega_1, \mathbb{R})}$. Then in $V[g]$:

b. $V^\omega \subseteq V$, $\mathbb{R} = \mathbb{R}^V$, $\omega_1 = \omega_1^V$, $|\mathbb{R}| = \omega_1$;

c. $\Theta = \Theta^V = \omega_2$

d. $V[g] = L(\mathcal{P}(\mathbb{R}))$

e. ω_1 -DC holds;

f. $\text{Col}(\omega_2, \mathcal{P}(\mathbb{R}))$ is ω_1 -closed and homogeneous.

Let $h \rightsquigarrow V[g]^{\text{Col}(\omega_2, \mathcal{P}(\mathbb{R}))}$. Then in $V[g][h]$:

g. ZFC holds;

h. $V[g]^{\omega_1} \subseteq V[g]$, $V^\omega \subseteq V$, $\mathbb{R} = \mathbb{R}^V$, $\omega_1 = \omega_1^V$, $|\mathbb{R}| = \omega_1$.

²A poset $\mathbb{P}$ is *homogeneous* if for every $p, q \in \mathbb{P}$, there are $p' \leq p$ and $q' \leq q$ such that there exists an isomorphism $\sigma : \mathbb{P} \restriction p' \cong \mathbb{P} \restriction q'$. If $\mathbb{P}$ is homogeneous, $p \in \mathbb{P}$, and $p \Vdash \phi(\bar{x})$, then $\Vdash \phi(\bar{x})$.

- i. $|\mathcal{P}(\mathbb{R})| = \omega_2 = \Theta^V$.

Proof.

Part c.

- (1) Obviously, $\Theta^V \leq \Theta$.
- (2) Assume towards contradiction that $\Theta^V < \Theta$. Then there is a surjection $f : \mathbb{R} \rightarrow \Theta^V$.
- (3) In V , let $\dot{f}$ be such that $\Vdash \dot{f} : \mathbb{R} \rightarrow \Theta^V$ and let

$$F : \text{Col}(\omega_1, \mathbb{R}) \times \mathbb{R} \rightarrow \Theta$$

be defined by

$$F(p, x) := \begin{cases} \alpha, & p \Vdash \dot{f}(x) = \alpha, \\ 0, & \text{otherwise.} \end{cases}$$

- (4) Clearly, F is a surjection.
- (5) Continuing in V , there is a surjection $h : \mathbb{R} \rightarrow \text{Col}(\omega_1, \mathbb{R}) \times \mathbb{R}$, so $F \circ h : \mathbb{R} \rightarrow \Theta$ is a surjection. This is a contradiction.
- (6) Thus, $\Theta = \Theta^V$. Since $\Theta > \omega_1$ is a cardinal, we have $\Theta \geq \omega_2$.
- (7) For $\alpha < \Theta$, we have a surjection $f : \mathbb{R} \rightarrow \alpha$ (or $\alpha = 0$). Since $|\mathbb{R}| = \omega_1$, we conclude $|\alpha| \leq \omega_1$ and consequently $\alpha < \omega_2$.
- (8) Thus, $\Theta \leq \omega_2$, concluding the argument.

Part e.

- (9)

□

[4.2.4] **Lemma.** Let $A \subseteq \mathbb{R}$ and suppose that N is a transitive model satisfying:

- a. $N \models \text{ZF}^- + \text{DC}$
- b. $\mathbb{R} \subseteq N$ and $A \in N$;
- c. $N \models (A \text{ is } \text{Col}(\omega, \mathbb{R})\text{-universally Baire})$.

Then for every countable $P \prec N$ with $A \in N$, for the transitive collapse $(\bar{P}, \in, \bar{A})$ of $(P, \in, A)$, and for every $h \rightsquigarrow \bar{P}^{\text{Col}(\omega, \mathbb{R})}$, we have $\bar{A}_h = A \cap \bar{P}[h]$.

Proof.

- (1) Let $j : \bar{P} \prec N$ be the anti-collapse and let $T, S \in \bar{P}$ witness $\text{Col}(\omega, \mathbb{R})$ -universal Baireness of $\bar{A}$ inside $\bar{P}$.
- (2) For $x \in \mathbb{R} \cap \bar{P}[h]$, we have

$$x \in \bar{A}_h \iff x \in p[T] \cap \bar{P}[h] \tag{5}$$

$$\implies x \in p[j(T)] \cap \bar{P}[h] \tag{6}$$

$$\iff x \in A \cap \bar{P}[h]; \tag{7}$$

(6) follows like this: if $(x, y) \in [T]$, then $((x \restriction n, j(y \restriction n)) : n < \omega)$ is a branch of $j(T)$.

- (3) Similarly,

$$x \in \mathbb{R}^{\bar{P}[h]} - \bar{A}_h \iff x \in p[S] \cap \bar{P}[h] \tag{8}$$

$$\implies x \in p[j(S)] \cap \bar{P}[h] \tag{9}$$

$$\iff x \in A^c \cap \bar{P}[h]. \tag{10}$$

- (4) This suffices for the conclusion $\bar{A}_h = A \cap \bar{P}[h]$.

□

[4.2.5] **Lemma.** Let $F : \mathbb{R} \rightarrow V$ be such that $F[\mathbb{R}]$ is a transitive model of $\text{ZF}^- + \text{DC}$ containing all reals and let $G \rightsquigarrow V^{\text{Col}(\omega_1, \mathbb{R})}$. Then there exists a club $D \subseteq [\mathbb{R}]^\omega$ in $V[G]$ such that for every $\sigma \in D$ and every $d \in \sigma$, if $F(d)$ is a dense subset of $\text{Col}(\omega_1, \mathbb{R})$, then there exists $x \in \sigma$ such that $F(x) \in F(d) \cap G$.

Proof.

- (1) Let $f : \mathbb{R}^{<\omega} \rightarrow \mathbb{R}$ satisfy: if $d \in \mathbb{R}$ and $F(d)$ is a dense subset of $\text{Col}(\omega_1, \mathbb{R})$, then $f((d)) \in \{x \in \mathbb{R} : F(x) \in F(d) \cap G\}$. The reason such choices can be made is because $\mathbb{R}$ can be well-ordered.
- (2) Set $D := C_f$ is as required.

□

[4.2.6] **Notation.**

- a. If $x \in \mathbb{R}$ codes a mapping $\alpha \rightarrow \mathbb{R}$ (with $\alpha < \omega_1$), then that mapping is denote by p_x .
- b. We fix notation $\mathbb{P} := \{x \in \mathbb{R} : p_x \text{ is defined}\} \subseteq \mathbb{R}$. We have the surjection

$$\mathbb{P} \rightarrow \text{Col}(\omega_1, \mathbb{R}) : x \mapsto p_x.$$

- c. Note that $\text{Col}(\omega_1, \mathbb{R})$ is a tree on $\mathbb{R}$. For any set N , we shall say that N is *tree-like* if

$$T_N := \{p_x : x \in \mathbb{P} \cap N\}$$

is a sub-tree³ of $\text{Col}(\omega_1, \mathbb{R})$. That tree is a γ -tree for some limit $\gamma \leq \omega_1$. If N is additionally countable, then γ can be chosen to be countable as well.

- d. If N is a transitive model of set theory, then N is tree-like and $T_N = \text{Col}(\omega_1, \mathbb{R})^N$.
- e. If σ is countable and tree like, then for every $g \in \mathcal{F}_{T_\sigma}$, we have that $p_g := \cup g : \alpha \rightarrow \mathbb{R}$ for some $\alpha < \omega_1$. In particular, $p_g \in \text{Col}(\omega_1, \mathbb{R})$.
- f. For $y \in \mathbb{P}$, we denote $\sigma_y := \text{im}(p_y) \in [\mathbb{R}]^\omega$. If we want to treat $\mathbb{P}$ as the set of codes for elements of $[\mathbb{R}]^\omega$, we denote it by Σ . We have surjection

$$\Sigma \rightarrow [\mathbb{R}]^\omega : y \mapsto \sigma_y.$$

[4.2.7] **Proposition.** Let $p \in \text{Col}(\omega_1, \mathbb{R})$ and let $\dot{S}$ satisfy $\text{Col}(\omega_1, \mathbb{R}) \Vdash \dot{S} \subseteq [\mathbb{R}]^\omega$. Then the following are equivalent:

- a. $p \Vdash_{\text{Col}(\omega_1, \mathbb{R})} (\dot{S} \text{ is a club})$
- b. for μ -a.e. $\sigma \in [\mathbb{R}]^\omega$ it holds that:
 - i. σ is tree-like;
 - ii. $p \in T_\sigma$
 - iii. set $\{g \in \mathcal{F}_{T_\sigma} : p_g \Vdash_{\text{Col}(\omega_1, \mathbb{R})} \sigma \in \dot{S}\}$ is co-meager in $\mathcal{N}_p^{T_\sigma}$.

Proof.

Implication (a $\Rightarrow$ b)

- (1) Let $\dot{f}$ be such that

$$\Vdash \dot{f} : \mathbb{R}^{<\omega} \rightarrow \mathbb{R}, \quad p \Vdash C_{\dot{f}} \subseteq \dot{S}$$

(where $C_{\dot{f}}$ is the club of closure points of $\dot{f}$).

- (2) **Claim.** There exists a club $C \subseteq [\mathbb{R}]^\omega$ such that for every $\sigma \in C$:

- a. σ is tree-like;
- b. $p \in T_\sigma$

³I.e. it is non-empty and closed for initial segments.

c. for every $t \in \sigma^{<\omega}$, set

$$D_t := \{q \in T_\sigma : (\exists x \in \sigma)(q \Vdash_{\text{Col}(\omega_1, \mathbb{R})} \dot{f}(t) = x)\}$$

is dense below p in T_σ .

Proof.

- (2.1) Whenever choices are made, they are possible due to $\text{AC}_\mathbb{R}$.
- (2.2) Let $F_0 : \mathbb{R}^{<\omega} \rightarrow \mathbb{R}$ satisfy: if $|s| > 0$ and $s(0)$ codes an ordinal α , then $F_0(s)$ codes a bijection $\omega \rightarrow \alpha$.
- (2.3) Let $F_1 : \mathbb{R}^{<\omega} \rightarrow \mathbb{R}$ satisfy: if $|s| > 0$ and if $s(0)$ codes a bijection $f : \omega \rightarrow \alpha$, then $F_1(s)$ codes $f(|s| - 1)$.
- (2.4) If $\sigma \in C_{F_0} \cap C_{F_1}$, then the set of ordinals coded by elements of σ is downward closed.
- (2.5) Let $F_2 : \mathbb{R}^{<\omega} \rightarrow \mathbb{R}$ satisfy: if $|s| > 0$ and $s(0) \in \mathbb{P}$, then $F_2(s)$ codes $\text{dom}(p_{s(0)})$.
- (2.6) Let $F_3 : \mathbb{R}^{<\omega} \rightarrow \mathbb{R}$ satisfy: if $|s| > 1$, if $s(0) \in \mathbb{P}$ with $p := p_{s(0)}$, and if $s(1)$ codes an $\beta < \text{dom}(p)$, then $F_3(s)$ codes $p \restriction \beta$.
- (2.7) If $\sigma \in C_{F_0} \cap C_{F_1} \cap C_{F_2} \cap C_{F_3}$, then T_σ is closed for initial segments.
- (2.8) Let $x_0 \in \mathbb{P}$ with $p_{x_0} = p$ and let $F_4 : \mathbb{R}^{<\omega} \rightarrow \mathbb{R} : s \mapsto x_0$. If $\sigma \in C_{F_0} \cap C_{F_1} \cap C_{F_2} \cap C_{F_3} \cap C_{F_4}$, then σ is tree-like and $p \in T_\sigma$.
- (2.9) Let $F_5 : \mathbb{R}^{<\omega} \rightarrow \mathbb{R}$ satisfy: if $|s| > 0$, if $s(0) \in \mathbb{P}$ with $p := p_{s(0)}$, and if t denotes the tuple s after removing the 0th coordinate, then $F_5(s) \in \mathbb{P}$ is such that $q := p_{F_5(s)} \leq p$ and $q \parallel \dot{f}(t)$.
- (2.10) Let $F_6 : \mathbb{R}^{<\omega} \rightarrow \mathbb{R}$ satisfy: if $|s| > 0$, if $s(0) \in \mathbb{P}$ with $p := p_{s(0)}$, if t denotes the tuple s after removing the 0th coordinate, and if $p \parallel \dot{f}(t)$, then $x := F_6(s)$ is such that $p \Vdash \dot{f}(t) = x$.
- (2.11) The club set $\cap \{C_{F_i} : i < 7\}$ is as required.

□

- (3) We will show that C is as required. Let $\sigma \in C$ be arbitrary. We have that σ is tree-like and that p belongs to the corresponding tree.
- (4) By [1.2.9], set

$$F := \bigcap_{t \in \sigma^{<\omega}} U_{D_t}$$

is co-meager in $\mathcal{N}_p$.

- (5) Let $g \in F$. We want to show $p_g \Vdash \sigma \in \dot{S}$, for which it suffices to show $p_g \Vdash \sigma \in C_{\dot{f}}$.
- (6) Let $t \in \sigma^{<\omega}$ be arbitrary and let us verify that $p_g \Vdash \dot{f}(t) \in \sigma$.
- (7) Since $g \in U_{D_t}$, there is $r \in D_t$ with $r \in g$. In particular, $p_g \leq r$.
- (8) Since $r \in D_t$, there exists $x \in \sigma$ such that $r \Vdash \dot{f}(t) = x$.
- (9) Hence, $p_g \Vdash \dot{f}(t) = x \in \sigma$, as required.

Implication (b $\Rightarrow$ a)

- (1) Let $A := \{(x, y) \in \mathbb{P} \times \Sigma : p_x \Vdash \sigma_y \in \dot{S}\}$.
- (2) By [4.1.6], there exists surjection $F : \mathbb{R} \rightarrow N$ such that N is a transitive model of $\text{ZF}^- + \text{DC}$ containing all reals and set A and satisfying

$$N \models (A \text{ is } \text{Col}(\omega, \mathbb{R})\text{-universally Baire}).$$

- (3) There exists a club $C \subseteq [\mathbb{R}]^\omega$ such that for every $\sigma \in C$:
 - a. σ is tree-like;
 - b. $p \in T_\sigma$
 - c. set $\{g \in \mathcal{F}_{T_\sigma} : p_g \Vdash_{\text{Col}(\omega_1, \mathbb{R})} \sigma \in \dot{S}\}$ is co-meager in $\mathcal{N}_p^{T_\sigma}$;
 - d. $F[\sigma] \prec N$ and $F[\sigma] \cap \mathbb{R} = \sigma$;
 - e. $A \in F[\sigma]$.

- (4) For $\sigma \in C$, let N_σ denote the collapse of $F[\sigma]$. We have $\mathbb{R}^{N_\sigma} = \sigma$ and $T_{N_\sigma} = T_\sigma = \text{Col}(\omega_1, \mathbb{R})^{N_\sigma}$. If $g \rightsquigarrow N_\sigma^{\text{Col}(\omega_1, \mathbb{R})}$, then $\mathbb{R}^{N_\sigma[g]} = \sigma$.
- (5) Let us write $\dot{g}$ for the canonical name for a $\text{Col}(\omega_1, \mathbb{R})$ -generic and $\dot{h}$ for the canonical name for a $\text{Col}(\omega, \mathbb{R})$ -generic. Poset $\text{Col}(\omega_1, \mathbb{R}) \times \text{Col}(\omega, \mathbb{R})$ is forcing-equivalent to $\text{Col}(\omega, \mathbb{R})$; let $\dot{g} \cdot \dot{h}$ be some canonical name for the $\text{Col}(\omega, \mathbb{R})$ -generic corresponding to $\dot{g} \times \dot{h}$.
- (6) **Claim.** For every $\sigma \in C$ and for $\bar{A}$ denoting the image of A under collapse $F[\sigma] \rightarrow N_\sigma$, we have

$$N_\sigma \models (p \Vdash_{\text{Col}(\omega_1, \mathbb{R})} (\Vdash_{\text{Col}(\omega, \mathbb{R})} (\forall x \in \mathbb{P})(\forall y \in \Sigma)(p_x = p_{\dot{g}} \wedge \sigma_y = \sigma \implies (x, y) \in \bar{A}_{\dot{g} \cdot \dot{h}}))).$$

Proof.

- (6.1) Assume otherwise. Then there exists $p' \in T_\sigma$ with $p' \leq p$ such that

$$N_\sigma \models (p' \Vdash_{\text{Col}(\omega_1, \mathbb{R})} (\nVdash_{\text{Col}(\omega, \mathbb{R})} (\forall x \in \mathbb{P})(\forall y \in \Sigma)(p_x = p_{\dot{g}} \wedge \sigma_y = \sigma \implies (x, y) \in \bar{A}_{\dot{g} \cdot \dot{h}}))).$$

- (6.2) Set $\{g \in \mathcal{F}_{T_\sigma} : p_g \Vdash_{\text{Col}(\omega_1, \mathbb{R})} \sigma \in \dot{S}\}$ is co-meager in $\mathcal{N}_p^{T_\sigma}$. Set $\{g \in \mathcal{F}_{T_\sigma} : g \rightsquigarrow N_\sigma^{\text{Col}(\omega_1, \mathbb{R})}\}$ is co-meager in $\mathcal{F}_{T_\sigma}$. In particular, their intersection is co-meager in $\mathcal{N}_p$.
- (6.3) Since $p' \leq p$, there exists $g \rightsquigarrow N_\sigma^{\text{Col}(\omega_1, \mathbb{R})}$ such that $p' \in g$ and $p_g \Vdash_{\text{Col}(\omega_1, \mathbb{R})} \sigma \in \dot{S}$.
- (6.4) We have

$$N_\sigma[g] \models (\nVdash_{\text{Col}(\omega, \mathbb{R})} (\forall x \in \mathbb{P})(\forall y \in \Sigma)(p_x = p_g \wedge \sigma_y = \sigma \implies (x, y) \in \bar{A}_{\dot{g} \cdot \dot{h}}))).$$

- (6.5) Let $h \rightsquigarrow N_\sigma[g]^{\text{Col}(\omega, \mathbb{R})}$ be such that

$$N_\sigma[g][h] \models (\exists x \in \mathbb{P})(\exists y \in \Sigma)(p_x = p_g \wedge \sigma_y = \sigma \wedge (x, y) \notin \bar{A}_{g \cdot h}).$$

- (6.6) Let $x \in \mathbb{P} \cap N_\sigma[g][h]$ and let $y \in \Sigma \cap N_\sigma[g][h]$ be such that $p_x = p_g$, $\sigma_y = \sigma$, and $(x, y) \notin \bar{A}_{g \cdot h}$.
- (6.7) By [4.2.4], we have $\bar{A}_{g \cdot h} = A \cap N_\sigma[g][h]$. Thus, $(x, y) \notin A$.
- (6.8) This means that $p_x \nVdash_{\text{Col}(\omega_1, \mathbb{R})} \sigma_y \in \dot{S}$. This is in a contradiction with (6.3) since $p_x = p_g$ and $\sigma_y = \sigma$.

□

- (7) Let $G \rightsquigarrow V^{\text{Col}(\omega_1, \mathbb{R})}$ with $p \in G$ be arbitrary. We want to find a club $D \subseteq [\mathbb{R}]^\omega$ such that $D \subseteq \dot{S}^G$.
- (8) By [4.2.5], there exists a club $D \subseteq C$ such that for $\sigma \in D$, filter $G \cap N_\sigma$ is $N_\sigma^{\text{Col}(\omega_1, \mathbb{R})}$ -generic. Let $\sigma \in D$ be arbitrary and let us verify $\sigma \in \dot{S}^G$.
- (9) Let $g := G \cap T_\sigma \rightsquigarrow N_\sigma^{\text{Col}(\omega_1, \mathbb{R})}$.
- (10) **Claim.** $p_g \in G$

Proof.

- (10.1) $\{q : q \leq p_g \vee q \perp p_g\} \in V$ is pre-dense, so there is $q \in G$ with $q \leq p_g$ or $q \perp p_g$.
- (10.2) Assume towards contradiction that $q \perp p_g$.
- (10.3) Then there is $\gamma < \text{dom}(p_g) = \omega_1^{N_\sigma}$ such that $(p_g \restriction \gamma) \perp q$.
- (10.4) Since $g \rightsquigarrow N_\sigma^{\text{Col}(\omega_1, \mathbb{R})}$, we have $p_g \restriction \gamma \in g \subseteq G$. This contradicts $q \in G$.

□

- (11) Take any $h \rightsquigarrow N_\sigma[g]^{\text{Col}(\omega, \mathbb{R})}$, let $x \in \mathbb{P} \cap N_\sigma[g][h]$ be such that $p_x = p_g$, and let $y \in \Sigma \cap N_\sigma[g][h]$ be such that $\sigma_y = \sigma$. By (6) and [4.2.4], we have $(x, y) \in \bar{A}_{g \cdot h} = A \cap N_\sigma[g][h]$.
- (12) Since $(x, y) \in A$, we conclude $p_x \Vdash_{\text{Col}(\omega_1, \mathbb{R})} \sigma_y \in \dot{S}$. Since $p_x = p_g \in G$ and $\sigma_y = \sigma$, this yields $\sigma \in \dot{S}^G$.

□

[4.2.8] **Corollary.** Let $p \in \text{Col}(\omega_1, \mathbb{R})$ and let $\dot{S}$ satisfy $\text{Col}(\omega_1, \mathbb{R}) \Vdash \dot{S} \subseteq [\mathbb{R}]^\omega$. Then the following are equivalent:

- a. $p \Vdash_{\text{Col}(\omega_1, \mathbb{R})} (\dot{S} \text{ is a club})$
- b. for μ -a.e. $\sigma \in [\mathbb{R}]^\omega$ it holds that:
 - i. σ is tree-like;
 - ii. $p \in T_\sigma$
 - iii. set $\{g \in \mathcal{F}_{T_\sigma} : p \in g \wedge (\exists q \leq p_g)(q \Vdash_{\text{Col}(\omega_1, \mathbb{R})} \sigma \notin \dot{S})\}$ is not meager in $\mathcal{F}_{T_\sigma}$.

□

[4.2.9] **Proposition.** Let $G \rightsquigarrow V^{\text{Col}(\omega_1, \mathbb{R})}$. Then $\text{NS}_{[\mathbb{R}]^\omega}$ has the ordinal covering property w.r.t. V inside $V[G]$.

Proof.

- (1) Let $f : [\mathbb{R}]^\omega \rightarrow \text{Ord}$ and let $S \subseteq [\mathbb{R}]^\omega$ be stationary. We want to find function $f' \in ([\mathbb{R}]^\omega \rightarrow \text{Ord})^V$ such that $\{\sigma \in S : f(\sigma) = f'(\sigma)\}$ stationary.

- (2) Let $\dot{f}, \dot{S}$ satisfy

$$\text{Col}(\omega_1, \mathbb{R}) \Vdash^V \dot{f} : [\mathbb{R}]^\omega \rightarrow \text{Ord} \wedge (\dot{S} \text{ is stationary})$$

and $\dot{f}^G = f$ and $\dot{S}^G = S$.

- (3) **Claim.** Inside V , for every $p \in \text{Col}(\omega_1, \mathbb{R})$, there exist $r \leq p$ and function $f' : [\mathbb{R}]^\omega \rightarrow \text{Ord}$ such that

$$r \Vdash (\{\sigma \in \dot{S} : \dot{f}(\sigma) = f'(\sigma)\} \text{ is stationary}).$$

Proof.

- (3.1) $p \Vdash (\dot{S}^\mathbb{G} \text{ is club})$

- (3.2) By [4.2.8], for μ -a.e. $\sigma \in [\mathbb{R}]^\omega$, it holds that:

- a. σ is tree-like;
- b. $p \in T_\sigma$
- c. $\{g \in \mathcal{F}_{T_\sigma} : p \in g \wedge (\exists q \leq p_g)(q \Vdash \sigma \in \dot{S})\}$ is not meager in $\mathcal{F}_{T_\sigma}$.

Let $D \in \mu$ witness this fact.

- (3.3) Let $\sigma \in D$ be arbitrary. We have

$$\begin{aligned} & \{g \in \mathcal{F}_{T_\sigma} : p \in g \wedge (\exists q \leq p_g)(q \Vdash \sigma \in \dot{S})\} = \\ & \bigcup_{\alpha < \text{Ord}} \{g \in \mathcal{F}_{T_\sigma} : p \in g \wedge (\exists q \leq p_g)(q \Vdash (\sigma \in \dot{S} \wedge \dot{f}(\sigma) = \alpha))\}. \end{aligned}$$

Since the left side is not meager and AD holds, there exists α such that

$$\{g \in \mathcal{F}_{T_\sigma} : p \in g \wedge (\exists q \leq p_g)(q \Vdash (\sigma \in \dot{S} \wedge \dot{f}(\sigma) = \alpha))\}$$

is not meager. Let $f'(\sigma)$ be the minimal such α .

- (3.4) For every $\sigma \in D$, there exists $r \in T_\sigma$ such that $r \leq p$ and

$$\{g \in \mathcal{F}_{T_\sigma} : (\exists q \leq p_g)(q \Vdash (\sigma \in \dot{S} \wedge \dot{f}(\sigma) = f'(\sigma)))\}$$

is co-meager in $\mathcal{N}_r^{T_\sigma}$.

- (3.5) By normality, there exists $r \leq p$ such that for μ -a.e. $\sigma \in [\mathbb{R}]^\omega$, it holds that:

- a. σ is tree-like;
- b. $r \in T_\sigma$
- c. $\{g \in \mathcal{F}_{T_\sigma} : (\exists q \leq p_g)(q \Vdash (\sigma \in \dot{S} \wedge \dot{f}(\sigma) = f'(\sigma)))\}$ is co-meager in $\mathcal{N}_r$.

- (3.6) Hence, for every $s \leq r$, for μ -a.e. $\sigma \in [\mathbb{R}]^\omega$, it holds that:

- a. σ is tree-like;
- b. $s \in T_\sigma$
- c. set $\{g \in \mathcal{F}_{T_\sigma} : s \in g \wedge (\exists q \leq p_g)(q \Vdash (\sigma \in \dot{S} \wedge \dot{f}(\sigma) = f'(\sigma)))\}$ is not meager in $\mathcal{F}_{T_\sigma}$.

By [4.2.8], for every $s \leq r$, we have that

$$s \Vdash (\{\sigma \in [\mathbb{R}]^\omega : \sigma \notin \dot{S} \vee \dot{f}(\sigma) \neq f'(\sigma)\} \text{ is club}).$$

(3.7) Hence, $r \Vdash (\{\sigma \in \dot{S} : f(\sigma) = f'(\sigma)\} \text{ is stationary})$.

□

(4) Since G is V -generic, there exists $f' : [\mathbb{R}]^\omega \rightarrow \text{Ord}$ in V such that $\{\sigma \in S : f(\sigma) = f'(\sigma)\}$ is stationary.

□

[4.2.10] **Corollary.** Let $G \rightsquigarrow V^{\text{Col}(\omega_1, \mathbb{R})}$ and let $H \rightsquigarrow V[G]^{\text{Col}(\omega_2, \mathcal{P}(\mathbb{R}))}$. Then inside $V[G][H]$, $\text{NS}_{[\mathbb{R}]^\omega}$ has the ordinal covering property w.r.t. V .

Proof.

- (1) Let $f : [\mathbb{R}]^\omega \rightarrow \text{Ord}$ and let $S \subseteq [\mathbb{R}]^\omega$ be stationary. We want to find function $f' \in ([\mathbb{R}]^\omega \rightarrow \text{Ord})^V$ such that $\{\sigma \in S : f(\sigma) = f'(\sigma)\}$ stationary.
- (2) We recall that $([\mathbb{R}]^\omega)^V = ([\mathbb{R}]^\omega)^{V[G]} = ([\mathbb{R}]^\omega)^{V[G][H]}$ and note that $[\mathbb{R}]^\omega, S, f$ can be represented as the images of functions $\omega_1^{V[G]} \rightarrow V[G]$. The latter observation follows from the fact that $|\mathbb{R}|^{V[G]} = \omega_1^{V[G]}$.
- (3) Since $\text{Col}(\omega_2, \mathbb{R})$ is ω_1 -closed in $V[G]$ and since $V[G] \models \omega_1\text{-DC}$, we conclude that $f, S \in V[G]$ and that set S is stationary inside $V[G]$.
- (4) By the previous proposition, there exists $f' \in ([\mathbb{R}]^\omega \rightarrow \text{Ord}) \cap V$ such that $\{\sigma \in S : f(\sigma) = f'(\sigma)\}$ stationary in $V[G]$.
- (5) Since $[\mathbb{R}]^\omega$ is enumerated by ω_1 in $V[G]$ and since no sequence $\omega_1 \rightarrow V[G]$ is add in $V[G][H]$, we conclude that there are no new clubs in $[\mathbb{R}]^\omega$ inside $V[G][H]$. This means that $\{\sigma \in S : f(\sigma) = f'(\sigma)\}$ is stationary inside $V[G][H]$ as well.

□

[4.2.11] **Proposition.** Let $G \rightsquigarrow V^{\text{Col}(\omega_1, \mathbb{R})}$ and let $H \rightsquigarrow V[G]^{\text{Col}(\omega_2, \mathcal{P}(\mathbb{R}))}$. Let I be a normal ideal on $[\mathbb{R}]^\omega$ that has the ordinal covering property w.r.t. V inside $V[G][H]$ and let $K \rightsquigarrow V^{\mathbb{P}_I}$. Then the following holds:

- a. $K \cap V = \mu$
- b. mapping $k : \text{Ord} \rightarrow \text{Ord} : [f]_\mu \mapsto [f]_K$ ($f \in ([\mathbb{R}]^\omega \rightarrow \text{Ord}) \cap V$) is the identity;
- c. $\text{Ult}(V[G][H], K)$ is well-founded;
- d. $j_K \upharpoonright \text{Ord} = j_\mu \upharpoonright \text{Ord}$
- e. $j_K(\omega_1^{V[G][H]}) = \mathfrak{c}^{+V[G][H]}$.

Proof.

- (1) Inside V , every subset of $[\mathbb{R}]^\omega$ either contains a club or it is non-stationary. A function that witnesses either of these possibilities, witnesses the same fact in $V[G][H]$.
- (2) Since K is $V[G][H]$ -normal, it contains all club sets, so $K \cap V = \mu$, i.e. **a** holds.
- (3) This immediately implies that k is injective. For **b** it suffices then to verify that k is also surjective.
- (4) Let $f \in ([\mathbb{R}]^\omega \rightarrow \text{Ord}) \cap V[G][H]$ be arbitrary and let us find $g \in ([\mathbb{R}]^\omega \rightarrow \text{Ord}) \cap V$ such that $[f]_K = [g]_K$.
- (5) By the ordinal covering property, we have that

$$D := \{S \in \mathbb{P}_I : (\exists g \in ([\mathbb{R}]^\omega \rightarrow \text{Ord}) \cap V)(f \upharpoonright S = g \upharpoonright S)\}$$

is dense in $\mathbb{P}_I$.

- (6) Since $K \rightsquigarrow V^{\mathbb{P}_I}$, there is $S \in D \cap K$.
- (7) Let $g \in V$ witness $S \in D$. Then $S \in K$ witnesses $[f]_K = [g]_K$.

- (8) Since $k = \text{id}$ and $\text{Ult}(V, \mu)$ is well-founded, we see that the ordinals of $\text{Ult}(V[G][H], K)$ are well-founded. Thus, we can conclude [c](#).
- (9) Since $j_K \upharpoonright \text{Ord} = k \circ (j_\mu \upharpoonright \text{Ord})$ and $k = \text{id}$, we can conclude [d](#).
- (10) For [e](#), we have

$$j_K(\omega_1^{V[G][H]}) = j_K(\omega_1^V) \quad (11)$$

$$= j_\mu(\omega_1^V) \quad (12)$$

$$= \Theta^V \quad (13)$$

$$= \omega_2^{V[G][H]} \quad (14)$$

$$= \mathfrak{c}^{+V[G][H]}, \quad (15)$$

where (11), (14), (15) follow from [\[4.2.3\]](#), (12) follows from [d](#), and (13) follows from [\[2.0.10\]](#). $\square$

[4.2.12] Proposition. Let $G \rightsquigarrow V^{\text{Col}(\omega_1, \mathbb{R})}$, and let $H \rightsquigarrow V[G]^{\text{Col}(\omega_2, \mathcal{P}(\mathbb{R}))}$. Let I be a normal ideal on $[\mathbb{R}]^\omega$ inside $V[G][H]$ that has the ordinal covering property w.r.t. V . Then I is strong and pseudo-homogeneous.

Proof.

- (1) Strongness follows from [\[4.2.11\]](#). Let us verify pseudo-homogeneity.
- (2) Let $\alpha < \text{Ord}$, let $s \in \text{Ord}^\omega$, let $\lambda < \mathfrak{c}^+$, and let $\phi(x, y, z)$ be a formula. We want to verify that the truth value of the statement

$$\text{Ult}(V[G][H], \dot{K}) \models \phi[\alpha, j_{\dot{K}}(s), j_{\dot{K}}[\lambda^\omega]]$$

is decided by the weakest condition. (NB. that $\dot{K}$ denotes the canonical name for a $V[G][H]^{\mathbb{P}_I}$ -generic.)

- (3) There is $f \in ([\mathbb{R}]^\omega \rightarrow \text{Ord}) \cap V$ such that $\alpha = [f]_\mu$. By [\[4.2.11\]](#), we have $\Vdash \alpha = [f]_{\dot{K}}$.
- (4) Since $V^\omega \subseteq V$, it holds that $s \in V$.
- (5) We may assume w.l.o.g. that $\lambda > 0$. Since $\lambda < \mathfrak{c}^+ = \omega_2 = \Theta^V$, there exists surjection $g : \mathbb{R} \rightarrow \lambda^\omega$ in V . Let π be the function on $[\mathbb{R}]^\omega$ defined by $\pi(\sigma) := g[\sigma]$; function π is also in V and by [\[1.3.14\]](#)

$$V[G][H]^{\mathbb{P}_I} \models j_{\dot{K}}[\alpha^\omega] = [\pi]_{\dot{K}}.$$

(6)

$$\begin{aligned} & \|\text{Ult}(V[G][H], \dot{K}) \models \phi[\alpha, j_{\dot{K}}(s), j_{\dot{K}}[\lambda^\omega]]\|_{\mathbb{P}_I} = \\ & \|\{\sigma \in ([\mathbb{R}]^\omega)^{V[G][H]} : V[G][H] \models \phi[f(\sigma), s, \pi(\sigma)]\} \in \dot{K}\|_{\mathbb{P}_I} \end{aligned}$$

- (7) By homogeneity of the posets, it holds that

$$\{\sigma \in [\mathbb{R}]^\omega : V[G][H] \models \phi[f(\sigma), s, \pi(\sigma)]\} \quad (16)$$

$$= \{\sigma \in [\mathbb{R}]^\omega : V[G] \models (\text{Col}(\omega_2, \mathcal{P}(\mathbb{R})) \Vdash \phi[f(\sigma), s, \pi(\sigma)])\} \quad (17)$$

$$= \{\sigma \in [\mathbb{R}]^\omega : V \models (\text{Col}(\omega_1, \mathbb{R}) \Vdash (\text{Col}(\omega_2, \mathcal{P}(\mathbb{R})) \Vdash \phi[f(\sigma), s, \pi(\sigma)]))\} \in V. \quad (18)$$

- (8) By [\[4.2.11\]](#), it holds that $V[G][H]^{\mathbb{P}_I} \models \dot{K} \cap V = \mu$, so

$$\|\text{Ult}(V[G][H], \dot{K}) \models \phi[\alpha, j_{\dot{K}}(s), j_{\dot{K}}[\lambda^\omega]]\|_{\mathbb{P}_I}$$

is equal to the truth value of the statement

$$\{\sigma \in [\mathbb{R}]^\omega : V \models (\text{Col}(\omega_1, \mathbb{R}) \Vdash (\text{Col}(\omega_2, \mathcal{P}(\mathbb{R})) \Vdash \phi[f(\sigma), s, \pi(\sigma)]))\} \in \mu,$$

which is independent of a $V[G][H]^{\mathbb{P}_I}$ -generic. $\square$

[4.2.13] Corollary. Let $G \rightsquigarrow V^{\text{Col}(\omega_1, \mathbb{R})}$ and let $H \rightsquigarrow V[G]^{\text{Col}(\omega_2, \mathcal{P}(\mathbb{R}))}$. Then inside $V[G][H]$, ideal $\text{NS}_{[\mathbb{R}]^\omega}$ is strong and pseudo-homogeneous. $\square$

4.3 Forcing Dense Ideals

[4.3.1] **Summary.** We want to show that inside $(V^{\text{Col}(\omega_1, \mathbb{R})})^{\text{Col}(\omega_2, \mathcal{P}(\mathbb{R}))}$, there exists a normal $\mathfrak{c}$ -dense ideal on $[\mathbb{R}]^\omega$ with the ordinal covering property. Since $\mathfrak{c} = \omega_1$ in that model, we will be able to use this ideal to obtain a normal dense ideal I on ω_1 with the property that $j_{\dot{G}_{\mathbb{P}_I}} \upharpoonright \text{Ord}$ is decided by the weakest condition.

[4.3.2] **Notation.** For $G \rightsquigarrow V^{\text{Col}(\omega_1, \mathbb{R})}$ and $K \rightsquigarrow V[G]^{\text{Col}(\omega, \mathbb{R})}$, we denote by $G \cdot K$ the canonical generic for $V^{\text{Col}(\omega, \mathbb{R})}$ corresponding to $G * K$.

[4.3.3] **Lemma.** Let $G \rightsquigarrow V^{\text{Col}(\omega_1, \mathbb{R})}$, let $H \rightsquigarrow V[G]^{\text{Col}(\omega_2, \mathcal{P}(\mathbb{R}))}$, and let $K \rightsquigarrow V[G][H]^{\text{Col}(\omega, \mathbb{R})}$. There exists $G^* \rightsquigarrow \text{Ult}(V, U_{G \cdot K})^{\text{Col}(\omega_1, \mathbb{R})}$ such that $G \subseteq G^*$.

Proof.

- (1) Let $\mathfrak{M} := \text{Ult}(V, U_{G \cdot K})$ and let $\mathbb{Q} := \text{Col}(\omega_1, \mathbb{R})^{\mathfrak{M}}$.
- (2) By [3.2.9], we have $\mathbb{Q} = \text{Col}(\omega_1, \mathbb{R})^{V[G][K]}$.
- (3) We want to establish that there exists a surjection $\chi : \omega_1 \rightarrow \mathcal{P}(\mathbb{Q}) \cap \mathfrak{M}$ such that for every $\alpha < \omega_1$, it holds that $\chi \upharpoonright \alpha \in V[G][K]$.
- (3.1) Inside $V[G][K]$, there exists surjection

$$\chi_0 : (\mathbb{R}^\omega \rightarrow \mathcal{P}(\text{Col}(\omega_1, \mathbb{R})))^V \rightarrow \mathcal{P}(\mathbb{Q}) \cap \mathfrak{M}.$$

Indeed, if $A \in \mathcal{P}(\text{Col}(\omega_1, \mathbb{R}))^{\text{Ult}(V, U_{G \cdot K})}$, then there exists $f \in (\mathbb{R}^\omega \rightarrow \mathcal{P}(\text{Col}(\omega_1, \mathbb{R})))^V$ such that $A = [f]_{U_{G \cdot K}}$.

- (3.2) In V , there exists surjection $\chi_1 : \mathcal{P}(\mathbb{R}) \rightarrow (\mathbb{R}^\omega \rightarrow \mathcal{P}(\text{Col}(\omega_1, \mathbb{R})))^V$.
- (3.3) Hence, in $\chi_2 := \chi_0 \circ \chi_1 \in V[G][K]$ and

$$\chi_2 : \mathcal{P}(\mathbb{R})^V \rightarrow \mathcal{P}(\mathbb{Q}) \cap \mathfrak{M}$$

surjectively.

- (3.4) $\chi_3 := \bigcup H : \omega_2^{V[G]} \rightarrow \mathcal{P}(\mathbb{R})^{V[G]}$ surjectively and $\chi_3 \upharpoonright \alpha \in V[G]$ for every $\alpha < \omega_2^{V[G]}$.
- (3.5) Since $\omega_2^{V[G]} = \omega_2^{V[G][H]} = \omega_1^{V[G][H][K]} = \omega_1$, we conclude that there exists a surjection

$$\chi_4 : \omega_1 \rightarrow \mathcal{P}(\mathbb{R})^V$$

such that for every $\alpha < \omega_1$, it holds that $\chi_4 \upharpoonright \alpha \in V[G]$.

- (3.6) $\chi := \chi_2 \circ \chi_4$ is as required.
- (4) Since $G \subseteq \text{Col}(\omega_1, \mathbb{R})^V$, we conclude that $G \subseteq \mathbb{Q}$ and it is countable in $V[G][K]$.
- (5) In particular, there exists a lower bound $p_0 \in \mathbb{Q}$ of G .
- (6) By recursion, we define sequence $(p_\alpha : \alpha < \omega_1)$ of conditions in $\mathbb{Q}$ such that for every $\gamma < \omega_1$, it holds that $(p_\alpha : \alpha < \gamma) \in V[G][K]$.
- (6.1) Step $\alpha = 0$ is handled.
- (6.2) At recursive step $\alpha + 1$: if $\chi(\alpha)$ is dense in $\mathbb{Q}$, then set $p_{\alpha+1} \in \chi(\alpha)$ with $p_{\alpha+1} \leq p_\alpha$; otherwise, set $p_{\alpha+1} := p_\alpha$. The recursive hypothesis is preserved because $\chi \upharpoonright (\alpha + 1) \in V[G][K]$.
- (6.3) At recursive limit step γ , set $p_\gamma := \bigcup_{\gamma' < \gamma} p_{\gamma'} \in \mathbb{Q}$. The membership fact follows from the recursive hypothesis.
- (7) Filter $G^* := \{q \in \mathbb{Q} : (\exists \alpha < \omega_1)(p_\alpha \leq q)\}$ is as required.

□

[4.3.4] **Lemma.** Let $\mathbb{B}, \mathbb{C}$ be complete boolean algebras. Suppose that $\mathbb{B}$ is a complete sub-algebra of $\mathbb{C}$ and that $\mathbb{C}$ has a dense subset of cardinality $\leq \kappa$. Then $\mathbb{B}$ has a dense subset of cardinality $\leq \kappa$.

Proof.

- (1) Let $D \subseteq \mathbb{C}$ be dense of cardinality κ .
- (2) For $d \in D$, let $b_d := \prod \{b \in \mathbb{B} : b \geq d\} \in \mathbb{B} - \{0\}$.
- (3) Let $E := \{b_d : d \in D\} \subseteq \mathbb{B} - \{0\}$. Then E is dense in $\mathbb{B}$ and $|E| \leq \kappa$.

□

[4.3.5] **Proposition.** Let $G \rightsquigarrow V^{\text{Col}(\omega_1, \mathbb{R})}$ and let $H \rightsquigarrow V[G]^{\text{Col}(\omega_2, \mathcal{P}(\mathbb{R}))}$. Then inside $V[G][H]$, there exists $\mathfrak{c}$ -dense ideal I on $[\mathbb{R}]^\omega$ that has the ordinal covering property w.r.t. V .

Proof.

- (1) Let $\dot{F} \in V[G][H]^{\text{Col}(\omega, \mathbb{R})}$ have the following property: for every $K \rightsquigarrow V^{\text{Col}(\omega, \mathbb{R})}$, it holds that $\dot{F}^K \rightsquigarrow \text{Ult}(V, U_{G \cdot K})^{\text{Col}(\omega_1, \mathbb{R})}$ and $G \subseteq \dot{F}^K$ (cf. [4.3.3]).
- (2) Let $K \rightsquigarrow V^{\text{Col}(\omega, \mathbb{R})}$ and let $G^* := \dot{F}^K \rightsquigarrow V^{\text{Col}(\omega, \mathbb{R})}$. Mapping

$$k_K : V[G] \rightarrow \text{Ult}(V, U_{G \cdot K})[G^*] : \tau^G \mapsto j_{G \cdot K}(\tau)^{G^*}$$

is correctly defined and elementary and satisfies $k_K \upharpoonright V = j_{G \cdot K}$.

- (3) Inside $V[G][H]$, ideal I on $[\mathbb{R}]^\omega$ is defined by

$$S \in I \iff \text{Col}(\omega, \mathbb{R}) \Vdash \check{\mathbb{R}} \notin k_{\dot{K}}(\check{S}) \quad (\text{for all } S).$$

Note that we have used the fact that $\mathcal{P}([\mathbb{R}]^\omega) \subseteq V[G]$.

- (4) **Claim.** Ideal I is normal.

Proof.

- (4.1) Let us verify first that it is fine. Let $x \in \mathbb{R}$ and let us check that $\{S \in [\mathbb{R}]^\omega : x \notin S\} \in I$.

- (4.2) This amounts to verifying

$$\text{Col}(\omega, \mathbb{R}) \Vdash \check{\mathbb{R}} \notin k_{\dot{K}}(\{S : x \notin S\}),$$

which clearly holds since $\Vdash x \in \check{\mathbb{R}}$.

- (4.3) Let us now verify the closure for diagonal unions. Let $(S_x : x \in \mathbb{R}) \in I^{\mathbb{R}}$ be arbitrary and let us show that $\nabla_{x \in \mathbb{R}} S_x \in I$.

- (4.4) We need to show $\text{Col}(\omega, \mathbb{R}) \Vdash \check{\mathbb{R}} \notin k_{\dot{K}}(\nabla_{x \in \mathbb{R}} S_x)$. Let $K \rightsquigarrow V[G][H]^{\text{Col}(\omega, \mathbb{R})}$ be arbitrary and let us verify that

$$\mathbb{R}^V \notin k_K(\nabla_{x \in \mathbb{R}^V} S_x).$$

- (4.5) Let $f := (S_x : x \in \mathbb{R}^V)$. Since $V[G]$ is ω_1 -closed in $V[G][H]$ and $|\mathbb{R}|^{V[G]} = \omega_1^{V[G]}$, we have that $f \in V[G]$. Hence, our goal reduces to showing

$$\mathbb{R}^V \notin \nabla_{x \in \mathbb{R}} k_K(f)(x),$$

which is equivalent to

$$\mathbb{R}^V \notin \bigcup_{x \in \mathbb{R}^V} k_K(f)(x).$$

- (4.6) For every $x \in \mathbb{R}^V$, we have

$$k_K(f)(x) = k_K(f)(k_K(x)) = k_K(f(x)) = k_K(S_x) \not\supseteq \mathbb{R}^V,$$

where the last relation follows from the fact that $S_x \in I$. This yields the conclusion.

□

- (5) In particular, $\mathcal{P}([\mathbb{R}]^\omega)/I$ is $\mathfrak{c}$ -complete boolean algebra.

- (6) **Claim.** Mapping $i : \mathcal{P}([\mathbb{R}]^\omega)/I \rightarrow \mathbb{B}(\text{Col}(\omega, \mathbb{R})) : [S] \mapsto \|\check{\mathbb{R}} \in k_{\dot{K}}(\check{S})\|$ is a $\mathfrak{c}$ -complete monomorphsim.

Proof.

- (6.1) Mapping $i' : \mathcal{P}([\mathbb{R}]^\omega) \rightarrow \mathbb{B}(\text{Col}(\omega, \mathbb{R})) : S \mapsto \|\check{\mathbb{R}} \in k_{\check{K}}(\check{S})\|$ is clearly a homomorphism of boolean algebras. To show that i exists and it is monomorphism, we need to show that i' factors injectively through $\mathcal{P}([\mathbb{R}]^\omega) \twoheadrightarrow \mathcal{P}([\mathbb{R}]^\omega)/I$.

$$\begin{array}{ccc} \mathcal{P}([\mathbb{R}]^\omega) & \xrightarrow{i'} & \mathbb{B}(\text{Col}(\omega, \mathbb{R})) \\ \downarrow & \nearrow i & \\ \mathcal{P}([\mathbb{R}]^\omega)/I & & \end{array}$$

- (6.2) This reduces to showing that for every $S \subseteq [\mathbb{R}]^\omega$, it holds

$$S \in I \iff i'(S) = 0.$$

- (6.3) This is true since

$$i'(S) = 0 \iff \|\check{\mathbb{R}} \in k_{\check{K}}(\check{S})\| = 0 \iff \text{Col}(\omega, \mathbb{R}) \Vdash \check{\mathbb{R}} \notin k_{\check{K}}(\check{S}) \iff S \in I.$$

- (6.4) It remains to verify $\mathfrak{c}$ -completeness. To that end, let A be an anti-chain in $\mathcal{P}([\mathbb{R}]^\omega)/I$ of cardinality $\leq \mathfrak{c}$ and let us show that

$$i(\sum A) = \sum i[A].$$

- (6.5) Let us index A as $([S_x] : x \in \mathbb{R})$. We have

$$i(\sum A) = i(\sum_{x \in \mathbb{R}} [S_x]) = i([\bigvee_{x \in \mathbb{R}} S_x]) = \|\check{\mathbb{R}} \in k_{\check{K}}(\bigvee_{x \in \mathbb{R}} S_x)\|_{\mathbb{B}(\text{Col}(\omega, \mathbb{R}))}.$$

- (6.6) For a $K \rightsquigarrow V[G][H]^{\text{Col}(\omega, \mathbb{R})}$, let $\bar{K}$ denote the corresponding filter on $\mathbb{B}(\text{Col}(\omega, \mathbb{R}))$. Our problem reduces to showing that for every $K \rightsquigarrow V[G][H]^{\text{Col}(\omega, \mathbb{R})}$, we have

$$\|\check{\mathbb{R}} \in k_{\check{K}}(\bigvee_{x \in \mathbb{R}} S_x)\| \in \bar{K} \iff \sum i[A] \in \bar{K}.$$

- (6.7) This is justified as follows. Note that $f := (S_x : x \in \mathbb{R}^V) \in V[G]$.

$$\|\check{\mathbb{R}} \in k_{\check{K}}(\bigvee_{x \in \mathbb{R}} S_x)\| \in \bar{K} \iff \mathbb{R}^V \in k_K(\bigvee_{x \in \mathbb{R}^V} S_x) \quad (19)$$

$$\iff \mathbb{R}^V \in k_K(\nabla f) \quad (20)$$

$$\iff \mathbb{R}^V \in \nabla k_K(f) \quad (21)$$

$$\iff (\exists x \in \mathbb{R}^V)(\mathbb{R}^V \in k_K(f)(x)) \quad (22)$$

$$\iff (\exists x \in \mathbb{R}^V)(\mathbb{R}^V \in k_K(f(x))) \quad (23)$$

$$\iff (\exists x \in \mathbb{R}^V)(\mathbb{R}^V \in k_K(S_x)) \quad (24)$$

$$\iff (\exists x \in \mathbb{R}^V)(\|\check{\mathbb{R}} \in k_{\check{K}}((S_x)^\sim)\| \in \bar{K}) \quad (25)$$

$$\iff (\exists x \in \mathbb{R}^V)(i(S_x) \in \bar{K}) \quad (26)$$

$$\iff \sum i[A] \in \bar{K}. \quad (27)$$

□

- (7) **Claim.** Algebra $\mathcal{P}([\mathbb{R}]^\omega)/I$ is complete and monomorphism i is complete.

Proof.

- (7.1) Let A be an anti-chain in $\mathcal{P}([\mathbb{R}]^\omega)/I$. We want to show that $\sum A$ exists and $i(\sum A) = \sum i[A]$.
- (7.2) Image $i[A]$ is an anti-chain in $\mathbb{B}(\text{Col}(\omega, \mathbb{R}))$. Since $\mathbb{B}(\text{Col}(\omega, \mathbb{R}))$ has $\mathfrak{c}$ -c.c, we conclude that $|i[A]| \leq \mathfrak{c}$.
- (7.3) Consequently, $|A| \leq \mathfrak{c}$. Since $\mathcal{P}([\mathbb{R}]^\omega)/I$ is a $\mathfrak{c}$ -complete algebra and since i is a $\mathfrak{c}$ -complete homomorphism, the conclusion follows.

□

- (8) Since $\mathbb{B}(\text{Col}(\omega, \mathbb{R}))$ is $\mathfrak{c}$ -dense, lemma [4.3.4] implies that $\mathcal{P}([\mathbb{R}]^\omega)/I$ is $\mathfrak{c}$ -dense. In other words, I is $\mathfrak{c}$ -dense.
- (9) **Claim.** Ideal I has the ordinal covering property w.r.t. V .

Proof.

- (9.1) Let $S \in \mathbb{P}_I$ and let $f : [\mathbb{R}]^\omega \rightarrow \text{Ord}$. We want to find $f' \in ([\mathbb{R}]^\omega \rightarrow \text{Ord}) \cap V$ such that

$$\{\sigma \in S : f(\sigma) = f'(\sigma)\} \in \mathbb{P}_I.$$

- (9.2) Since $S \notin I$, there exists $p \in \text{Col}(\omega, \mathbb{R})$ such that $p \Vdash \check{\mathbb{R}} \in k_{\dot{K}}(\check{S})$.
- (9.3) We have $f \in V[G]$ and $\text{Col}(\omega, \mathbb{R}) \Vdash |\check{\mathbb{R}}| = \omega$. This means that $k_{\dot{K}}(f)(\check{\mathbb{R}})$ is a correctly defined ordinal in $V[G][H]^{\text{Col}(\omega, \mathbb{R})}$. Let $q \leq p$ and $\alpha \in \text{Ord}$ be such that $q \Vdash k_{\dot{K}}(f)(\check{\mathbb{R}}) = \alpha$.
- (9.4) There exists $f' : [\mathbb{R}]^\omega \rightarrow \text{Ord}$ inside V such that $\alpha = [f']_\mu$.
- (9.5) By proposition [3.2.8], we have

$$V[G][H]^{\text{Col}(\omega, \mathbb{R})} \models \alpha = [f']_\mu = j_{G, \dot{K}}(f')(\mathbb{R}^V) = k_{\dot{K}}(f')(\mathbb{R}^V).$$

- (9.6) Putting together (9.3) and (9.5), we conclude

$$(V[G][H], \text{Col}(\omega, \mathbb{R}), q) \Vdash k_{\dot{K}}(f)(\check{\mathbb{R}}) = k_{\dot{K}}(f')(\check{\mathbb{R}}).$$

- (9.7) Adding (9.2) to this, we obtain

$$(V[G][H], \text{Col}(\omega, \mathbb{R}), q) \Vdash \check{\mathbb{R}} \in k_{\dot{K}}(\{\sigma : f(\sigma) = f'(\sigma)\}) \cap k_{\dot{K}}(\check{S}),$$

i.e.

$$(V[G][H], \text{Col}(\omega, \mathbb{R}), q) \Vdash \check{\mathbb{R}} \in k_{\dot{K}}(\{\sigma \in S : f(\sigma) = f'(\sigma)\}).$$

- (9.8) This yields conclusion that $\{\sigma \in S : f(\sigma) = f'(\sigma)\} \notin I$.

□

□

[4.3.6] **Lemma.** Let $G \rightsquigarrow V^{\text{Col}(\omega_1, \mathbb{R})}$, and let $H \rightsquigarrow V[G]^{\text{Col}(\omega_2, \mathcal{P}(\mathbb{R}))}$. Let I be a normal ideal on $[\mathbb{R}]^\omega$ inside $V[G][H]$ that has the ordinal covering property w.r.t. V . Then I is predictable.

Proof. Proposition [4.2.12] implies that I is pseudo-homogeneous. Pseudo-homogeneity is a clear strengthening of predictability for ideals on $[\mathbb{R}]^\omega$. □

[4.3.7] **Corollary.** Let $G \rightsquigarrow V^{\text{Col}(\omega_1, \mathbb{R})}$ and let $H \rightsquigarrow V[G]^{\text{Col}(\omega_2, \mathcal{P}(\mathbb{R}))}$. Then inside $V[G][H]$, there exists a normal dense predictable ideal on ω_1 .

Proof.

- (1) Let I'' be a $\mathfrak{c}$ -dense ideal on $[\mathbb{R}]^\omega$ that has the ordinal covering property w.r.t. V (cf. [4.3.5]). In particular, ideal I'' is predictable by the previous lemma.
- (2) Since $|\mathbb{R}| = \omega_1$, we can obtain a normal ω_1 -dense predictable ideal I' on $[\omega_1]^\omega$.
- (3) Let $I := I' \cap \mathcal{P}(\omega_1)$. By [1.3.22] and [1.3.23], ideal I on ω_1 is normal, dense, and predictable.

□

[4.3.8] **Corollary.** If theory $\text{ZF} + \text{AD}_{\mathbb{R}} + (\Theta \text{ is regular})$ is consistent, then so is the theory $\text{ZFC} + \text{CH} + (\text{there exists a normal dense predictable ideal on } \omega_1)$.

Proof. This follows from [4.3.7] by recalling that CH holds in $V[G][H]$ (cf. [4.2.3]). □

References

References

- [Larson, 2019] Larson, P. (2019). Extensions of the axiom of determinacy. *To appear, available at <http://www.users.miamioh.edu/larsonpb>.*
- [Sargsyan et al., 2021] Sargsyan, G., Trang, N., and Wilson, T. (2021). Ideals and strong axioms of determinacy. *arXiv preprint [arXiv:2111.06220](https://arxiv.org/abs/2111.06220).*